

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

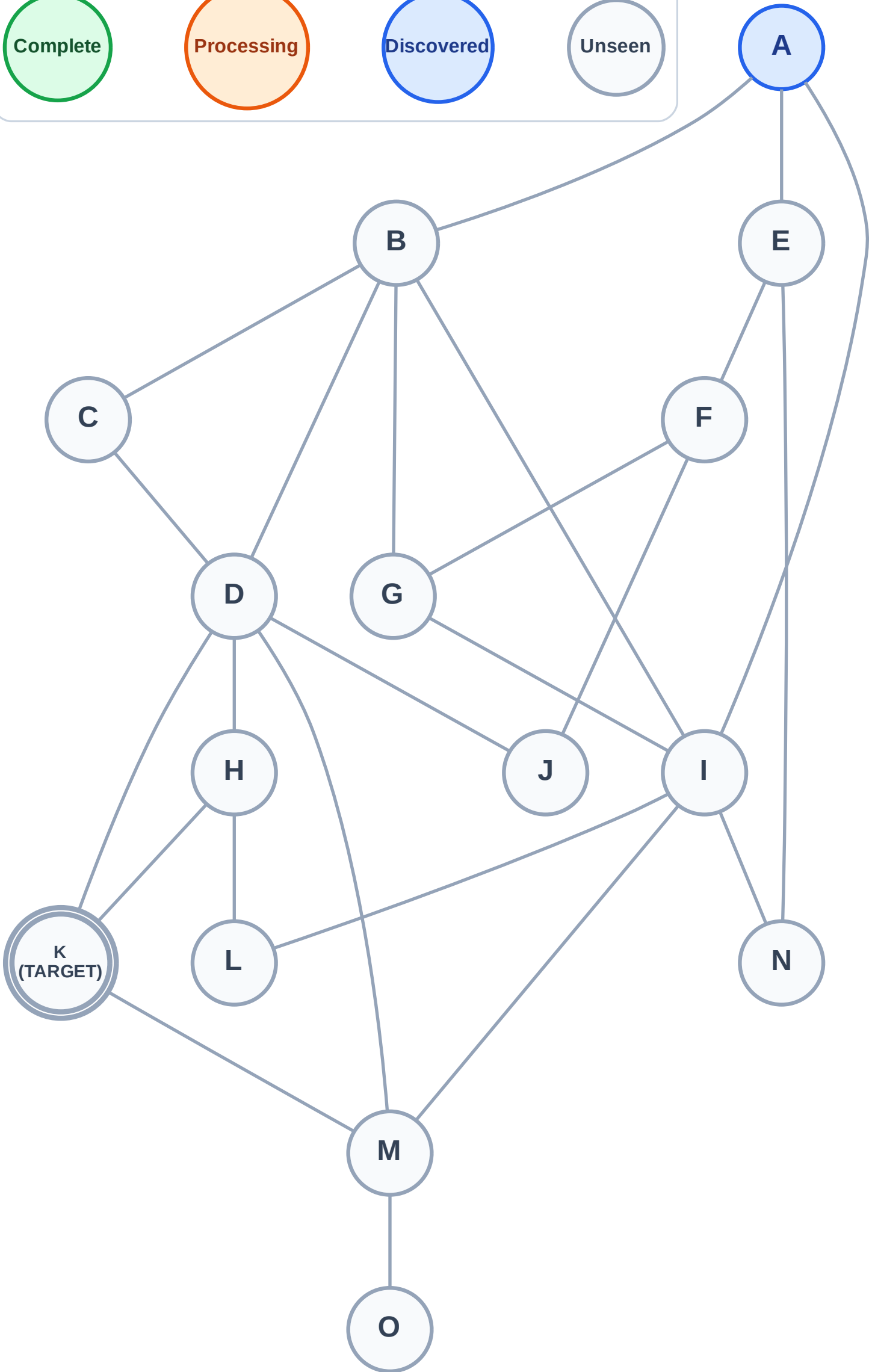
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

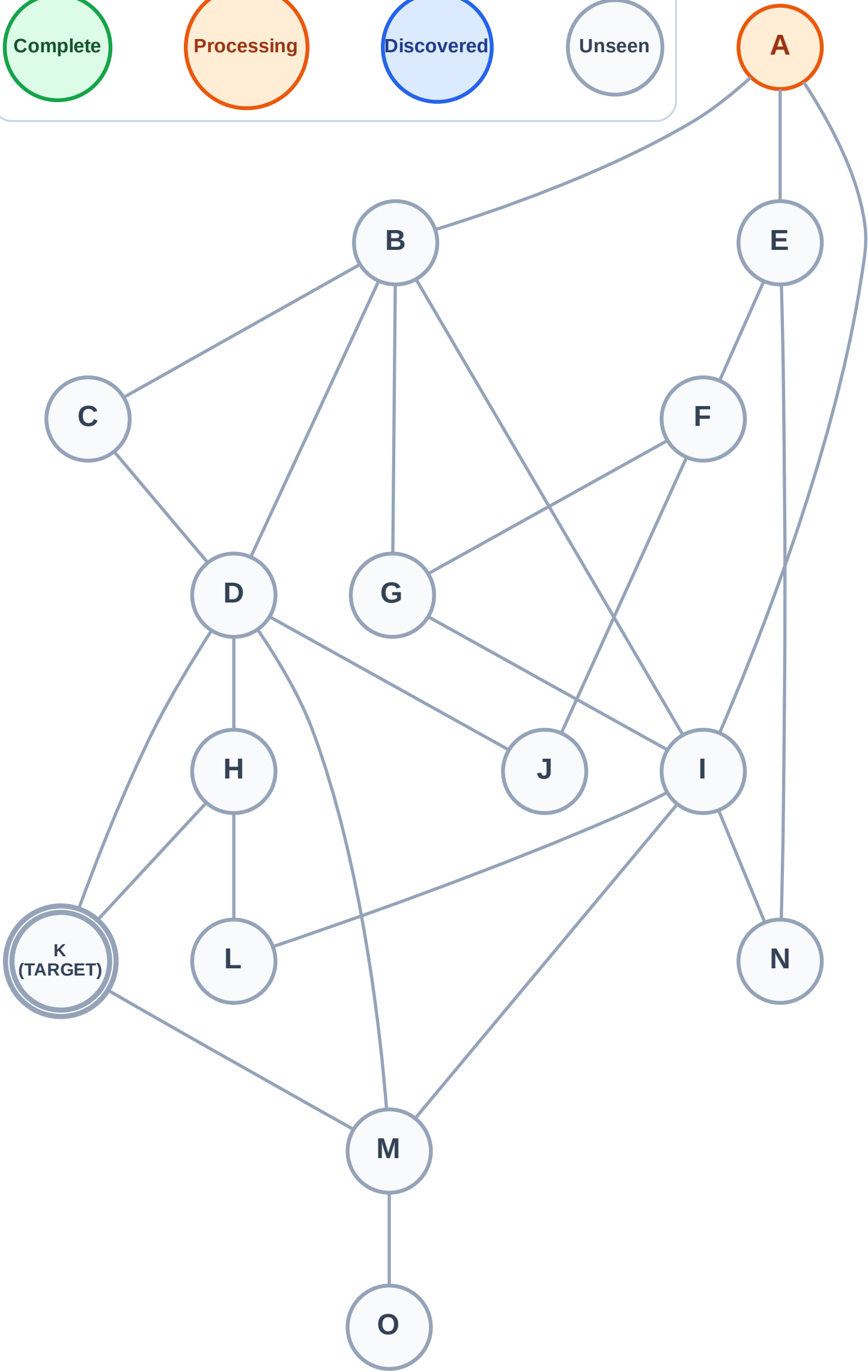
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, E, I from A

Legend

Complete

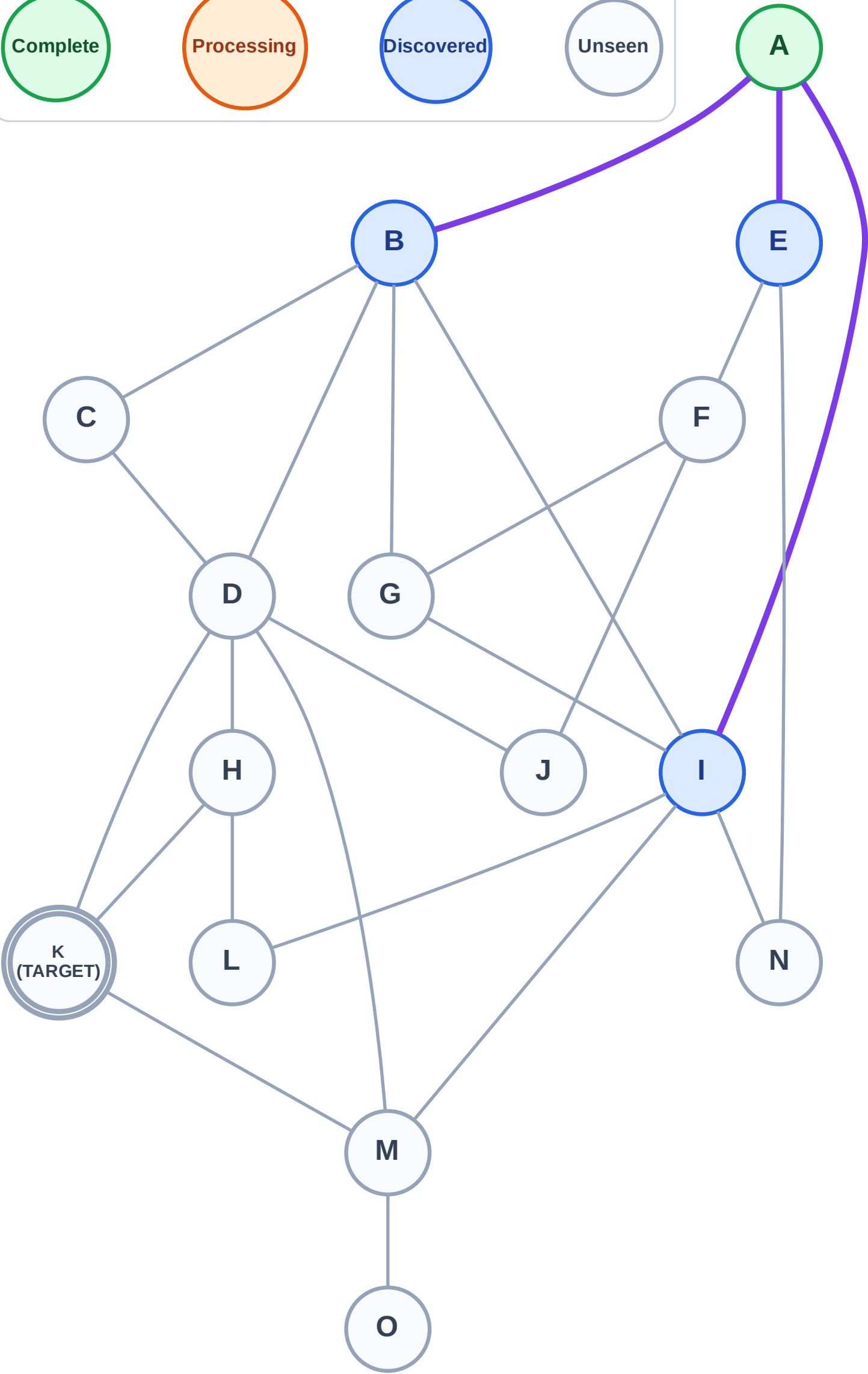
Processing

Discovered

Unseen

Queue: B → E → I

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

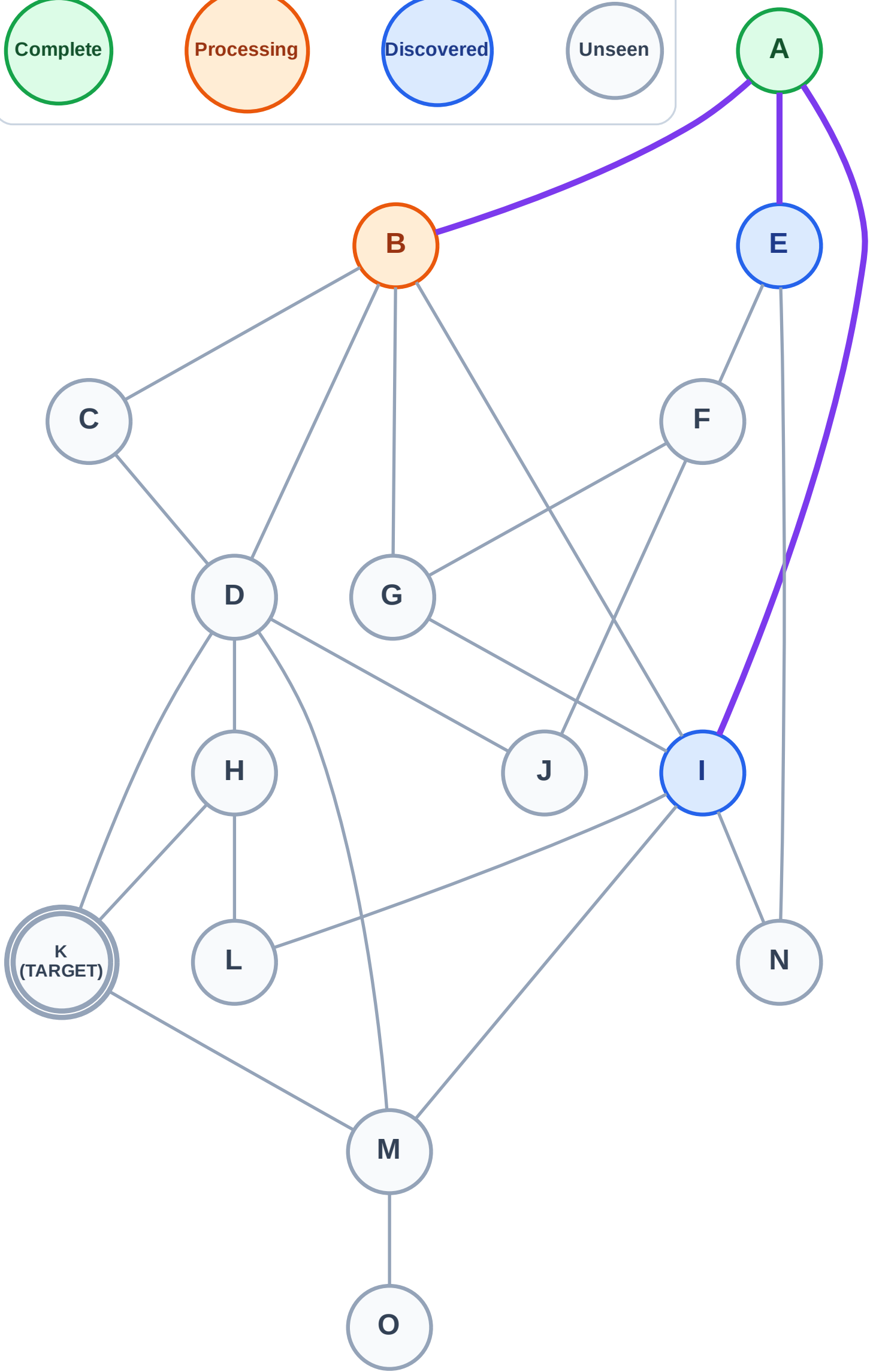
Processing

Discovered

Unseen

Queue: E → I

Visited: A



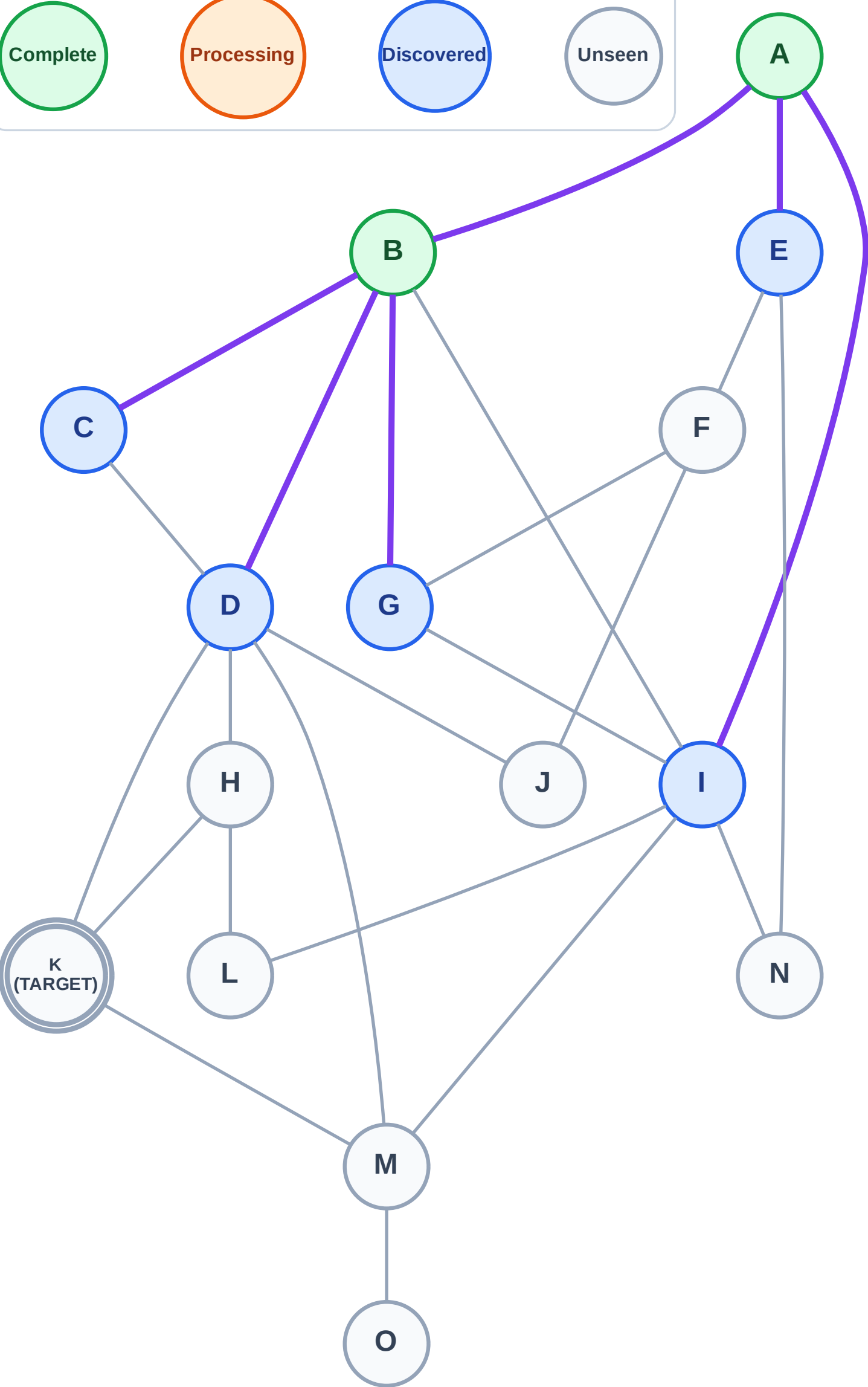
BFS Traversal

Step 5: Discover C, D, G from B

Legend



Queue: E → I → C → D → G
Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

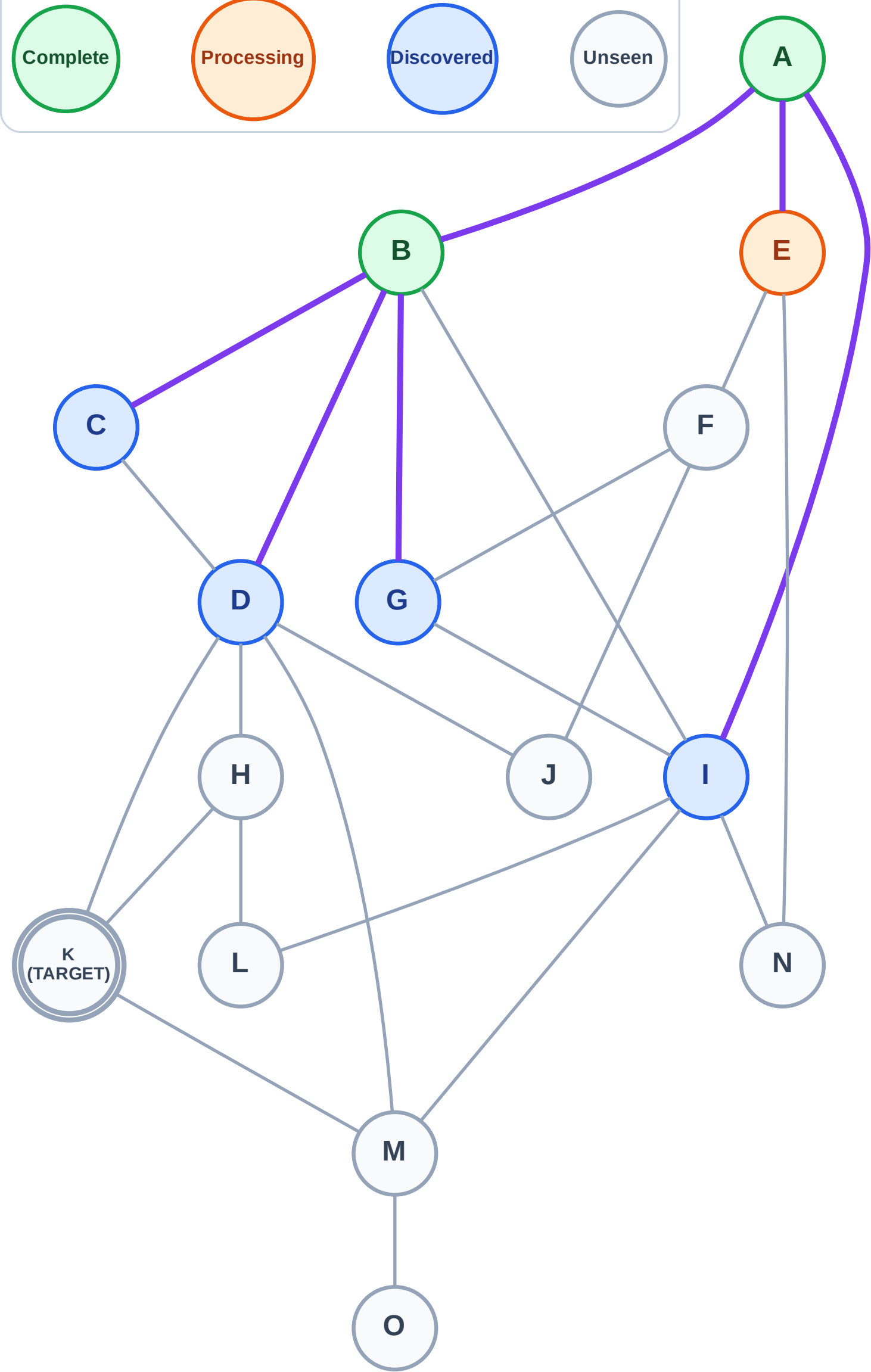
Processing

Discovered

Unseen

Queue: I → C → D → G

Visited: A, B



BFS Traversal

Step 7: Discover F, N from E

Legend

Complete

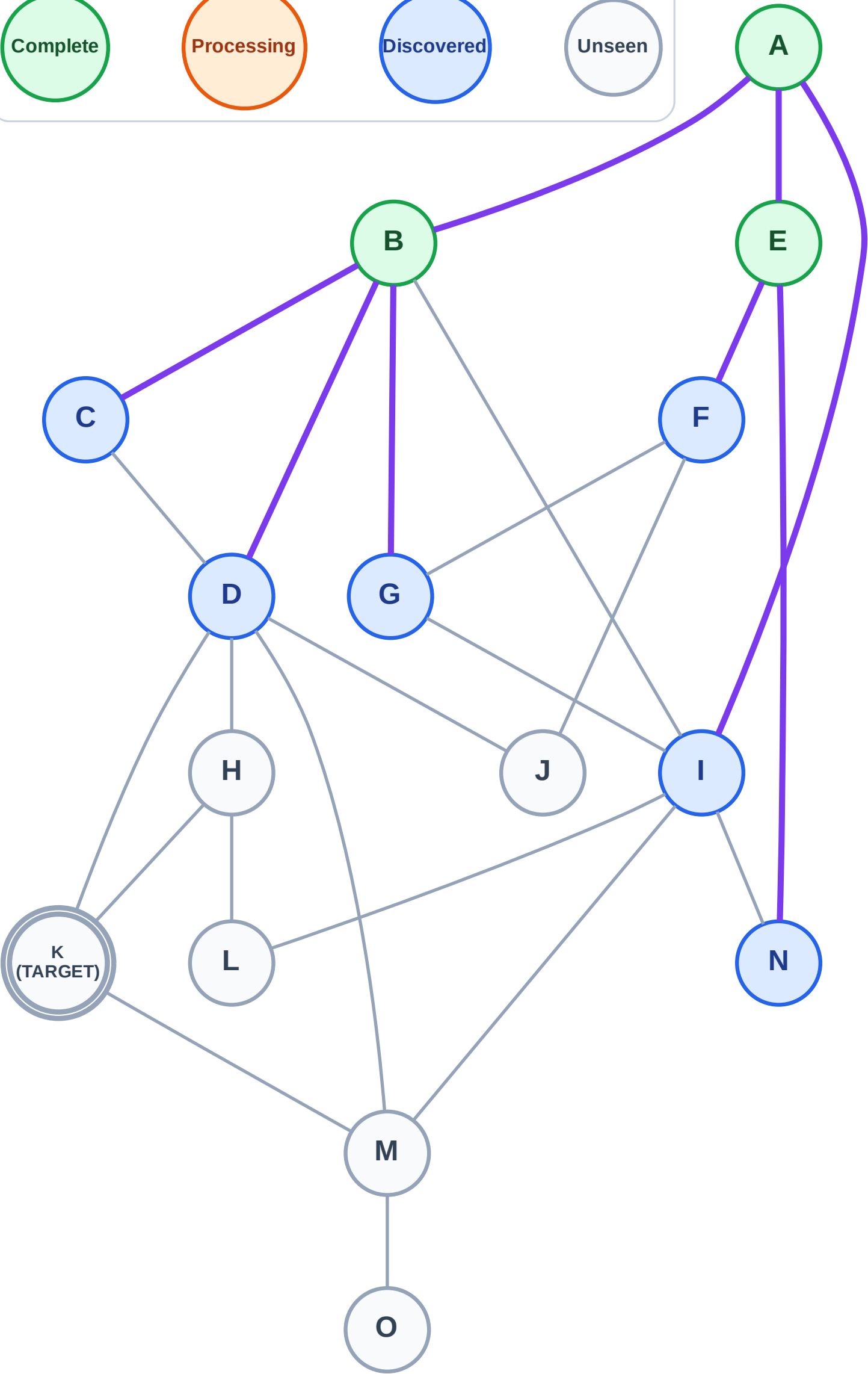
Processing

Discovered

Unseen

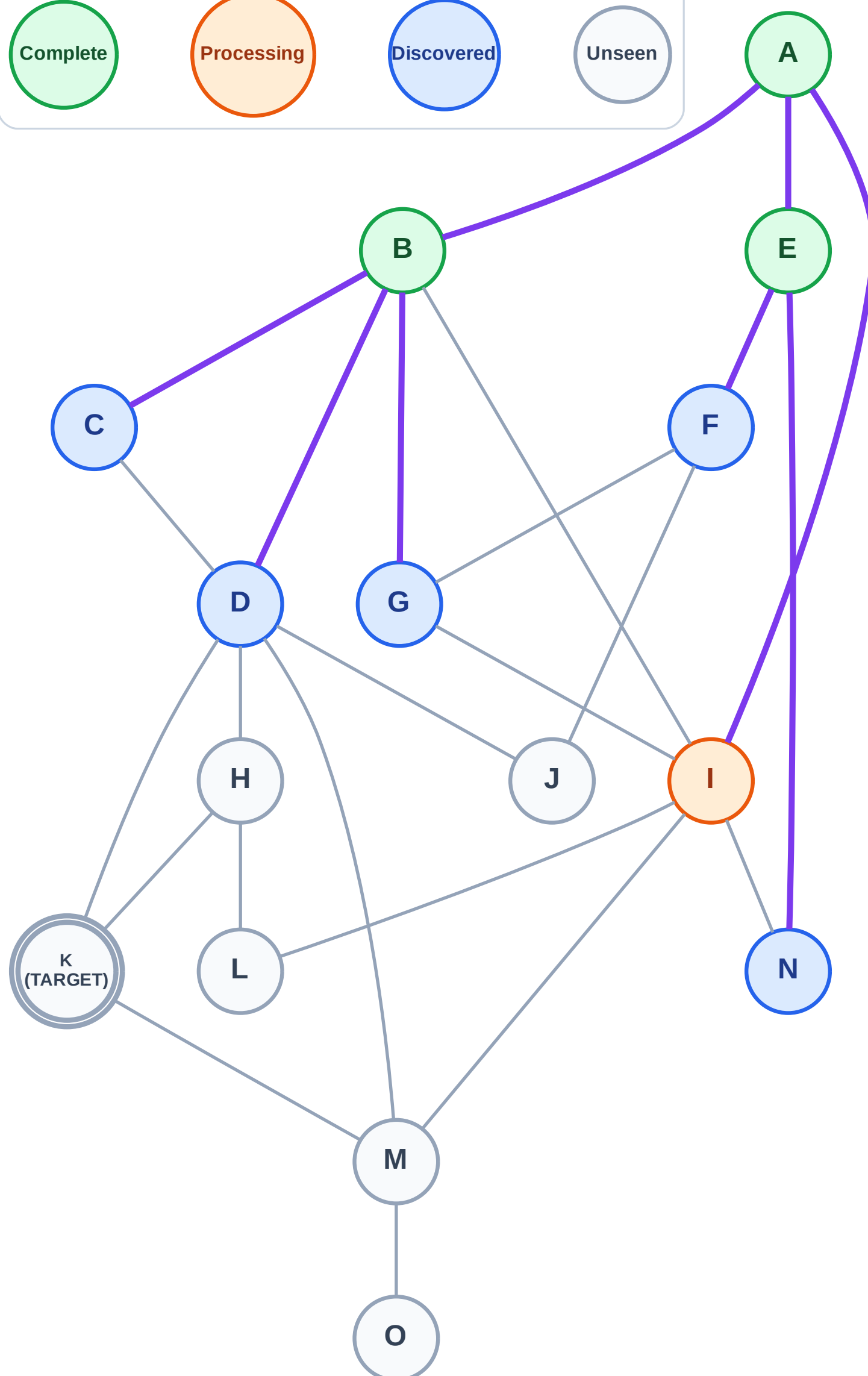
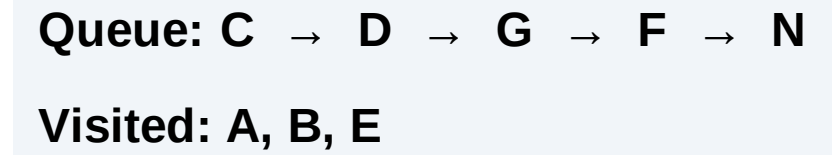
Queue: I → C → D → G → F → N

Visited: A, B, E



Step 8: Process node I

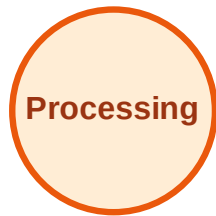
Step 8: Process node I



BFS Traversal

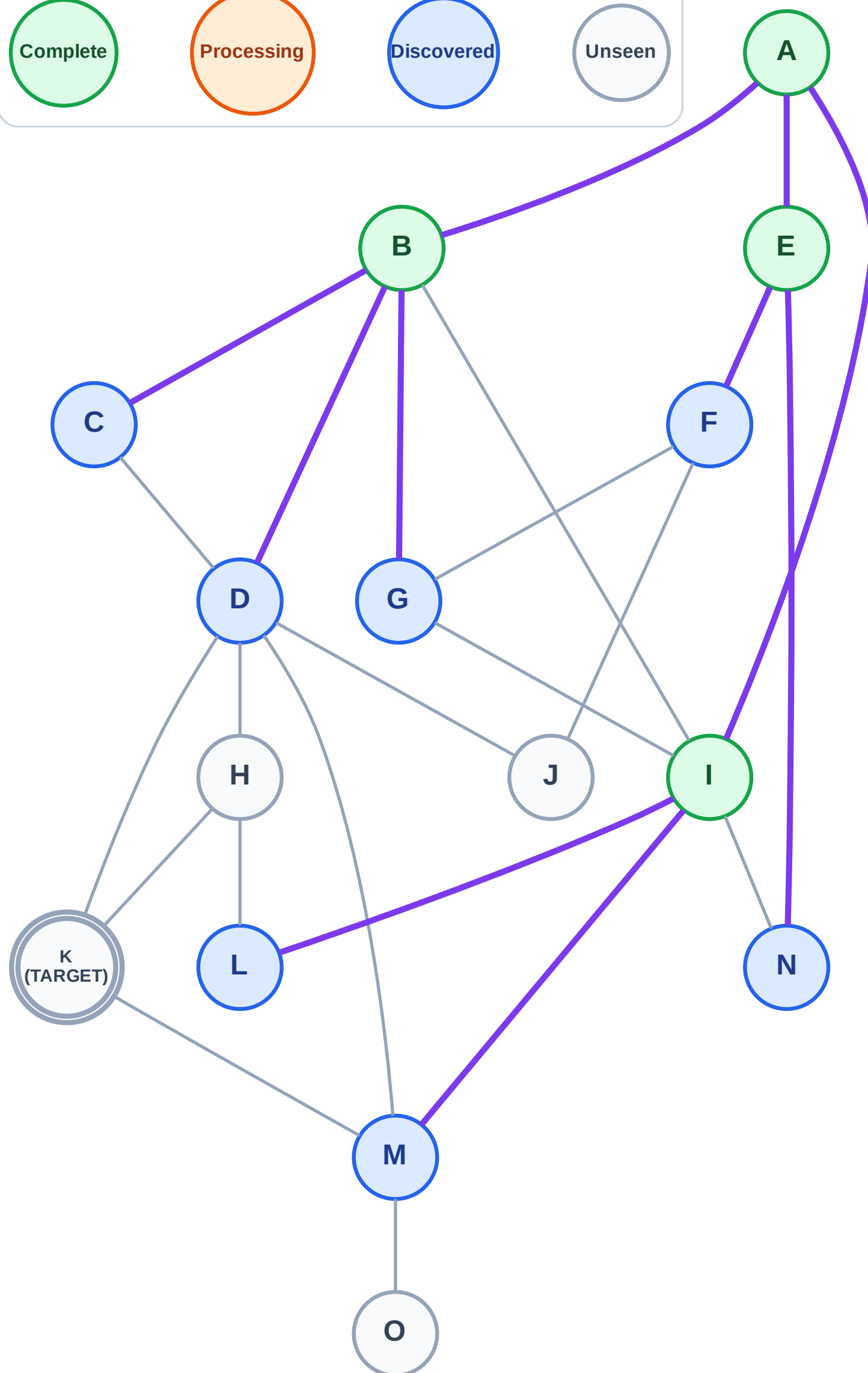
Step 9: Discover L, M from I

Legend



Queue: C → D → G → F → N → L → M

Visited: A, B, E, I



BFS Traversal

Step 10: Process node C

Legend

Complete

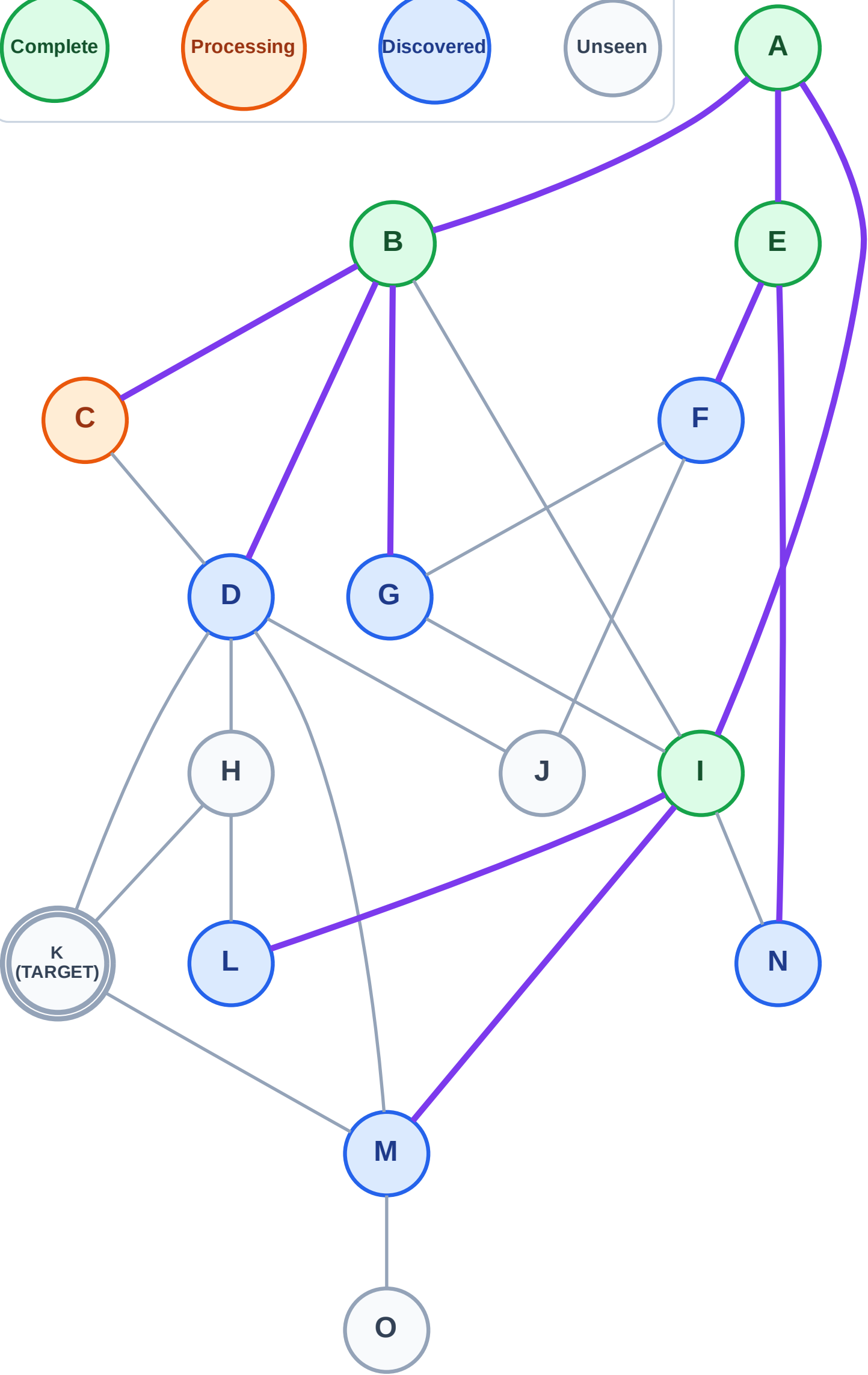
Processing

Discovered

Unseen

Queue: D → G → F → N → L → M

Visited: A, B, E, I



BFS Traversal

Step 11: No new neighbors from C

Legend

Complete

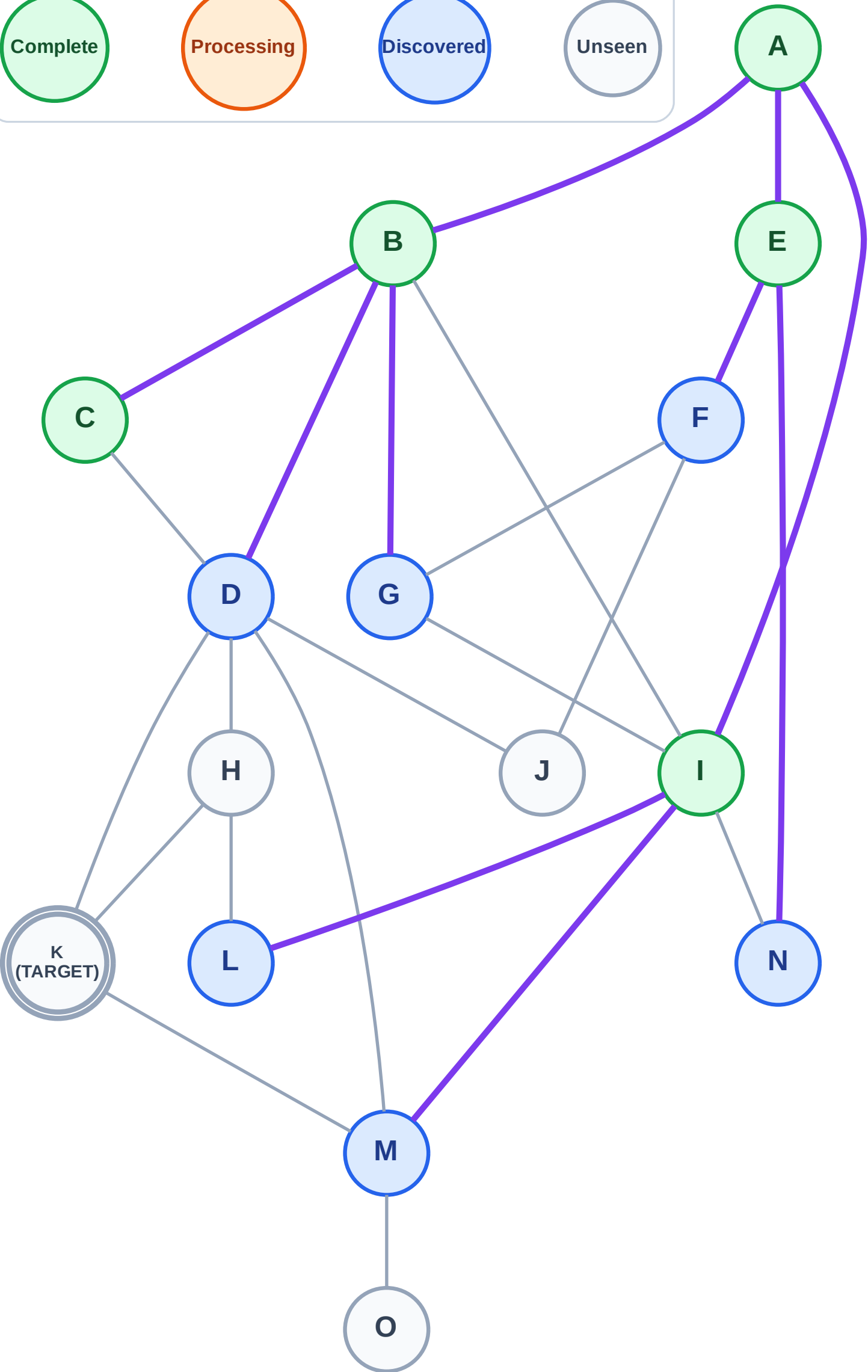
Processing

Discovered

Unseen

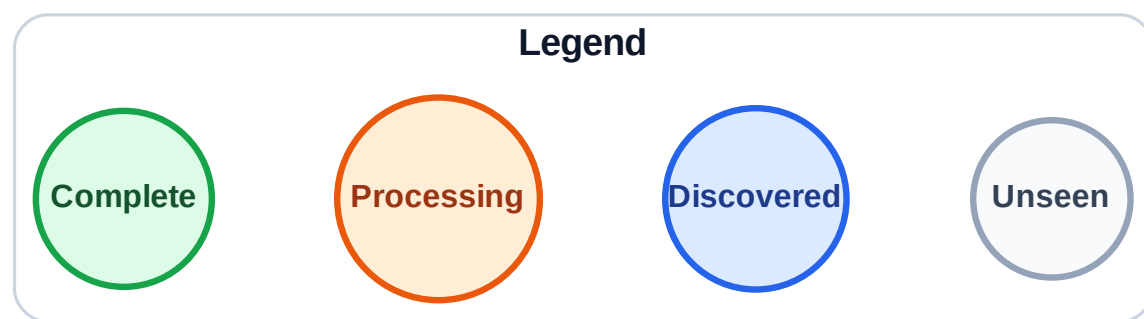
Queue: D → G → F → N → L → M

Visited: A, B, C, E, I



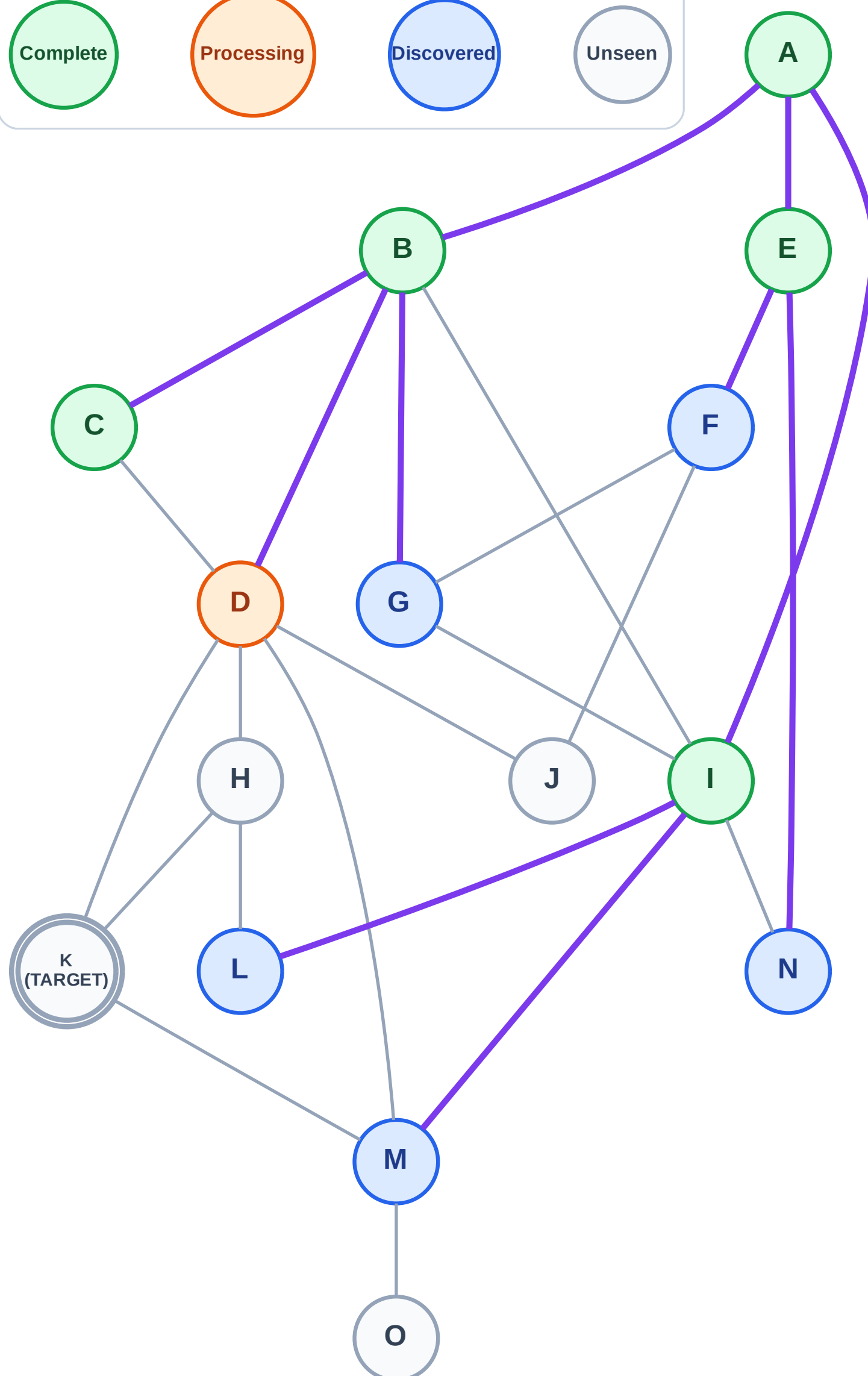
BFS Traversal

Step 12: Process node D



Queue: G → F → N → L → M

Visited: A, B, C, E, I



BFS Traversal

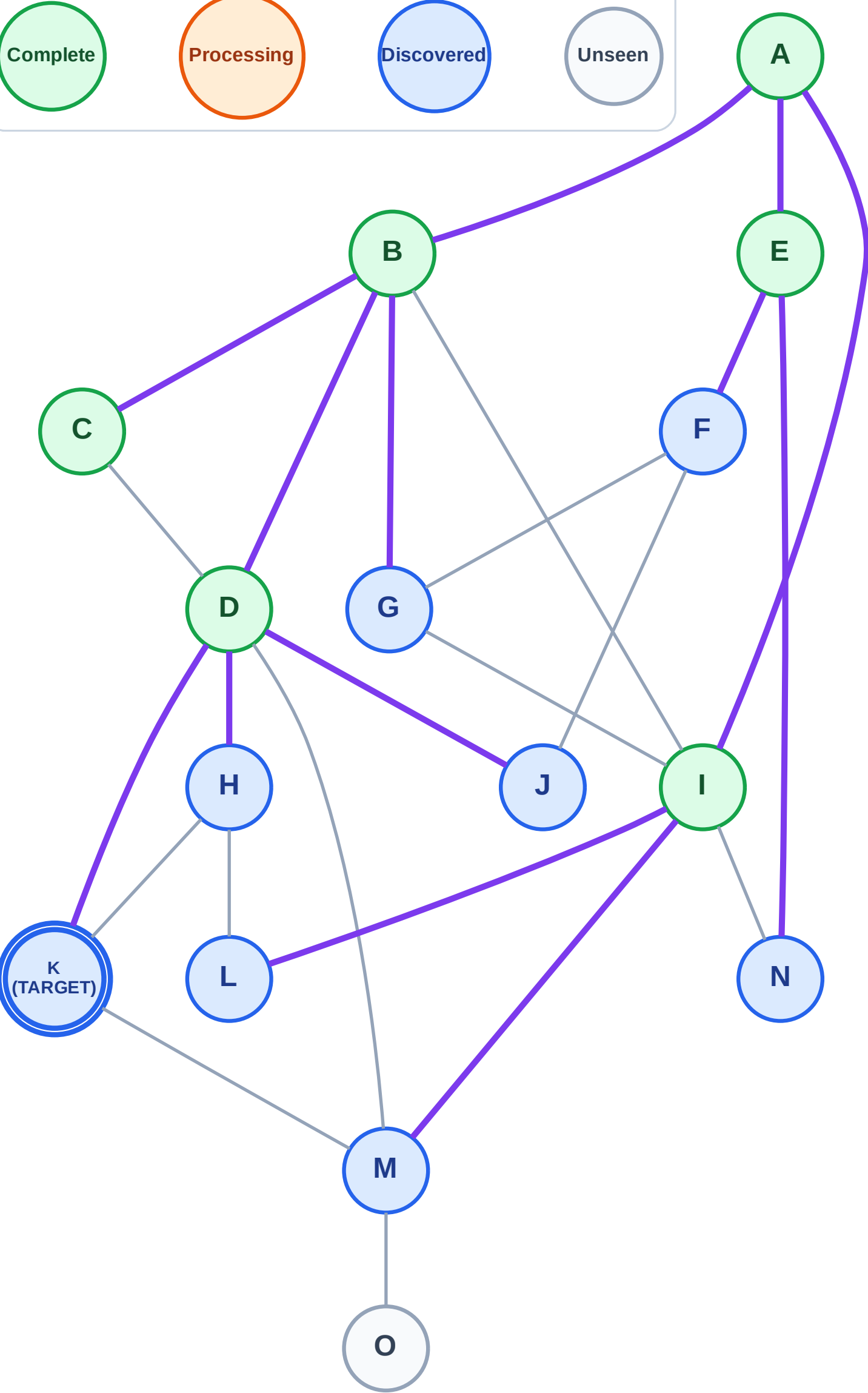
Step 13: Discover H, J, K from D

Legend



Queue: G → F → N → L → M → H → J → K

Visited: A, B, C, D, E, I



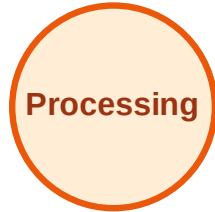
BFS Traversal

Step 14: Process node G

Legend



Complete



Processing



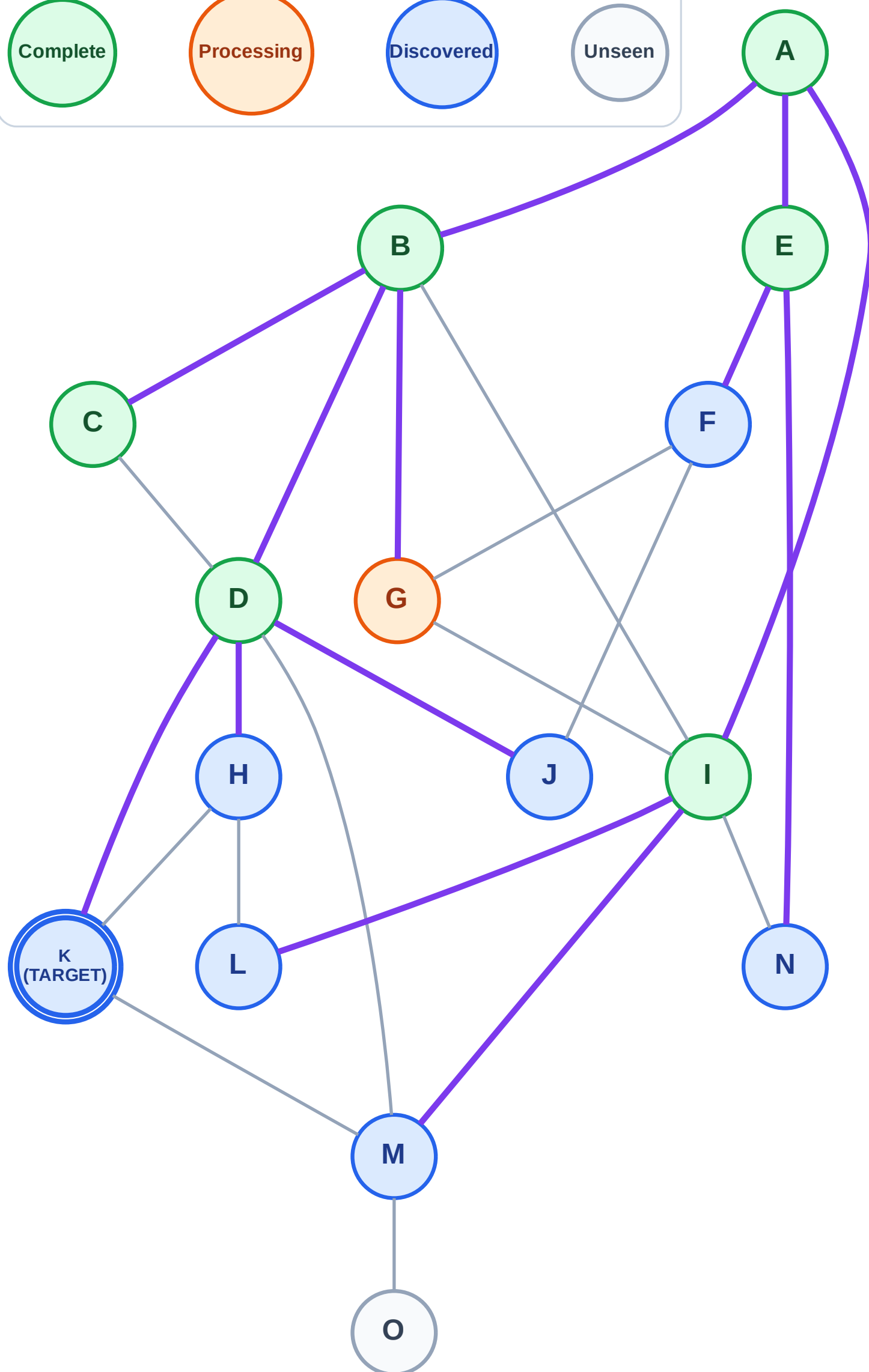
Discovered



Unseen

Queue: F → N → L → M → H → J → K

Visited: A, B, C, D, E, I



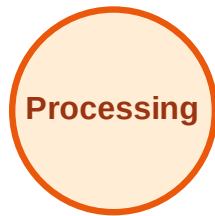
BFS Traversal

Step 15: No new neighbors from G

Legend



Complete



Processing



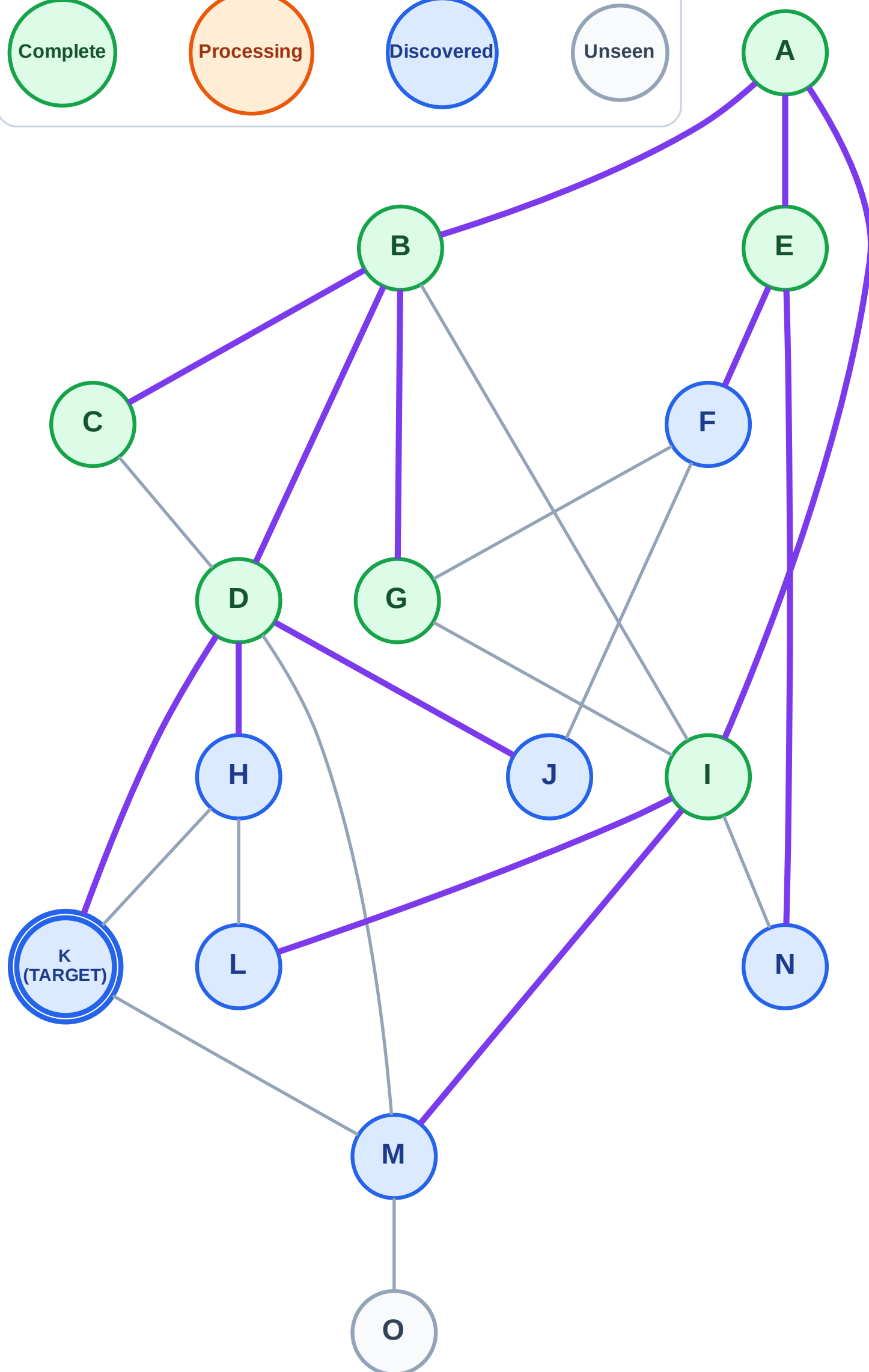
Discovered



Unseen

Queue: F → N → L → M → H → J → K

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node F

Legend

Complete

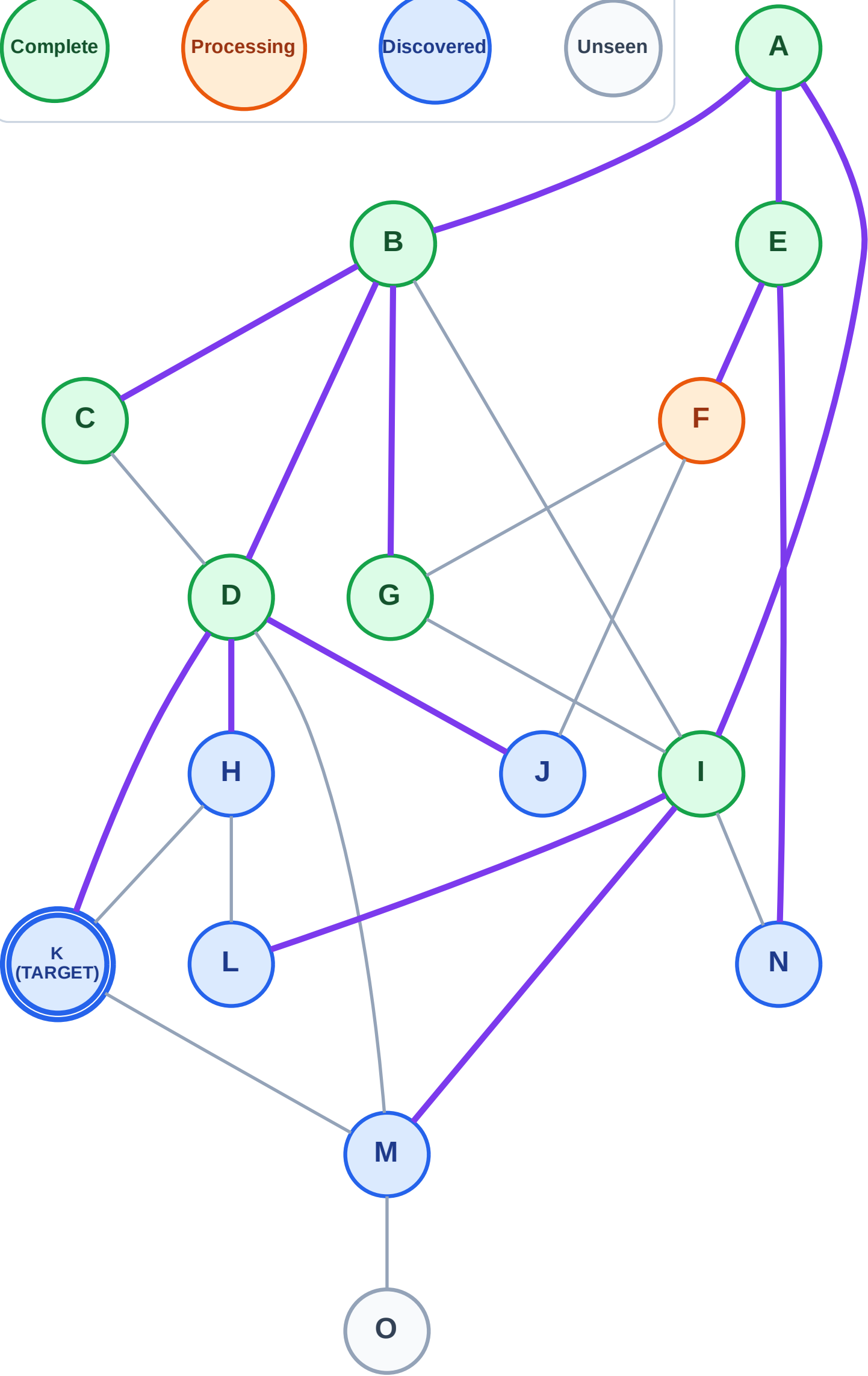
Processing

Discovered

Unseen

Queue: N → L → M → H → J → K

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 17: No new neighbors from F

Legend

Complete

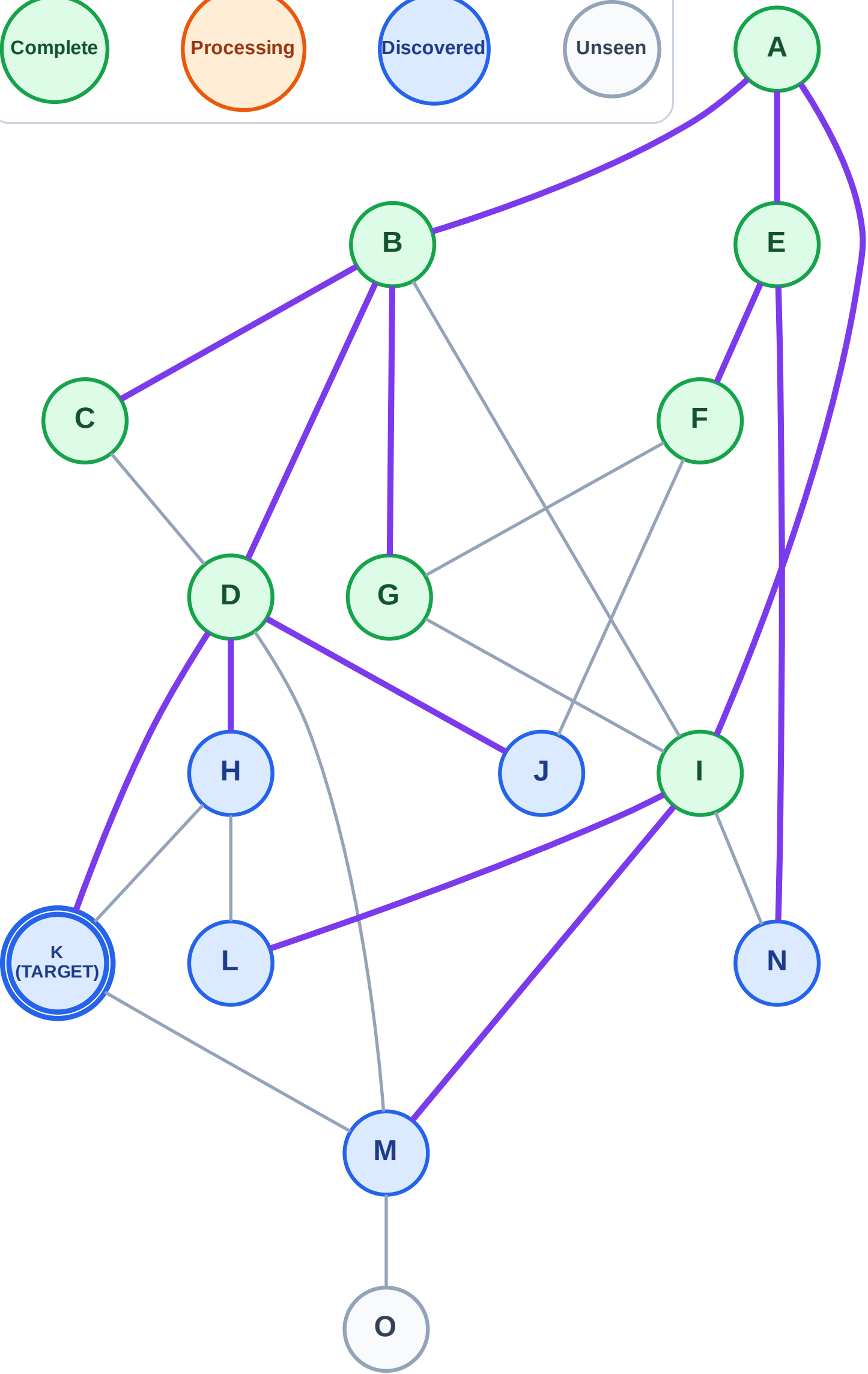
Processing

Discovered

Unseen

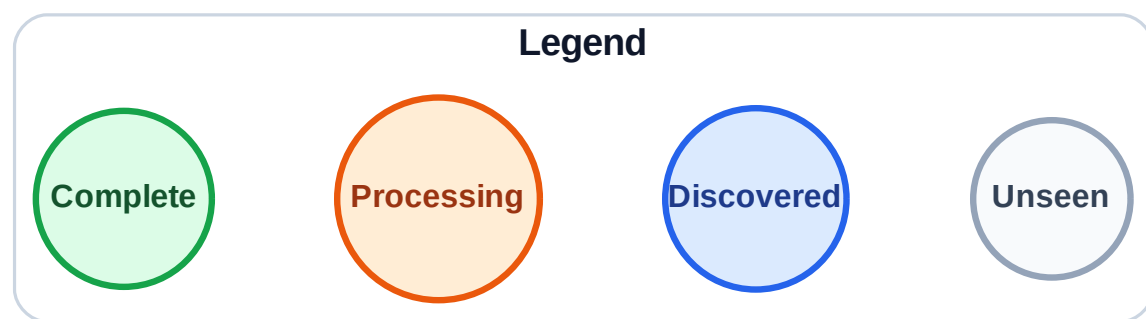
Queue: N → L → M → H → J → K

Visited: A, B, C, D, E, F, G, I



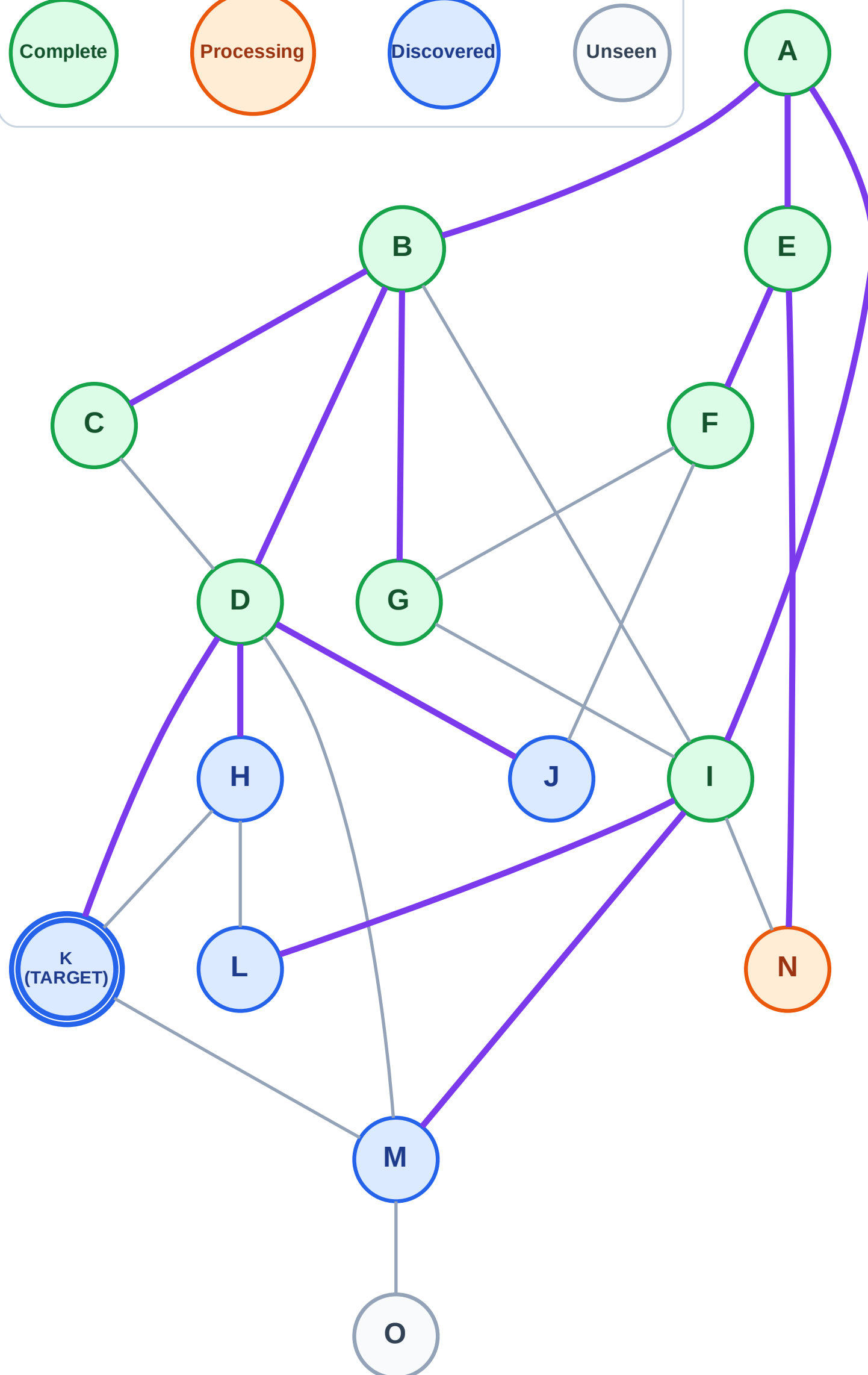
BFS Traversal

Step 18: Process node N



Queue: L → M → H → J → K

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 19: No new neighbors from N

Legend



Complete



Processing



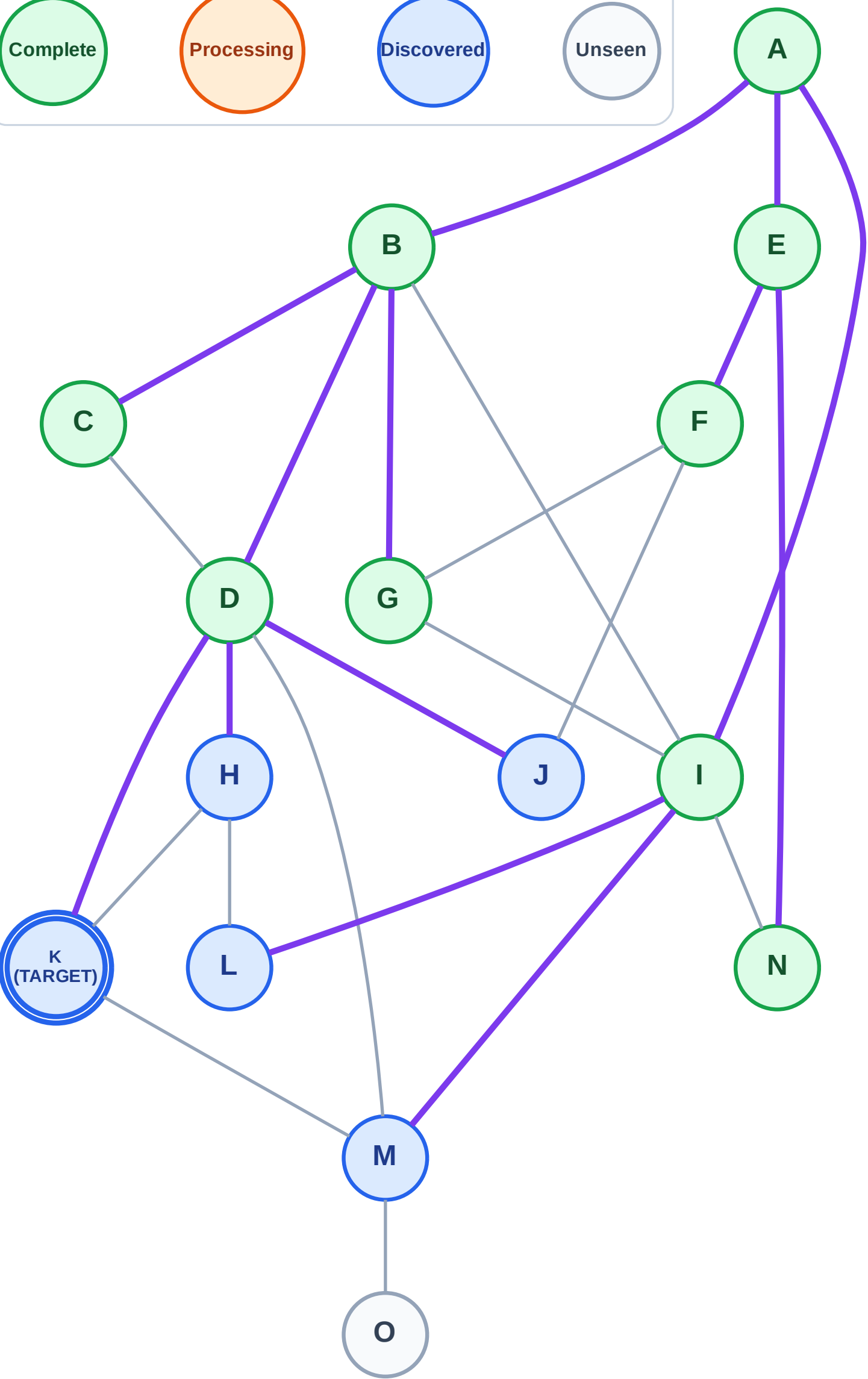
Discovered



Unseen

Queue: L → M → H → J → K

Visited: A, B, C, D, E, F, G, I, N



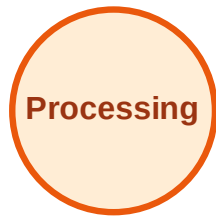
BFS Traversal

Step 20: Process node L

Legend



Complete



Processing



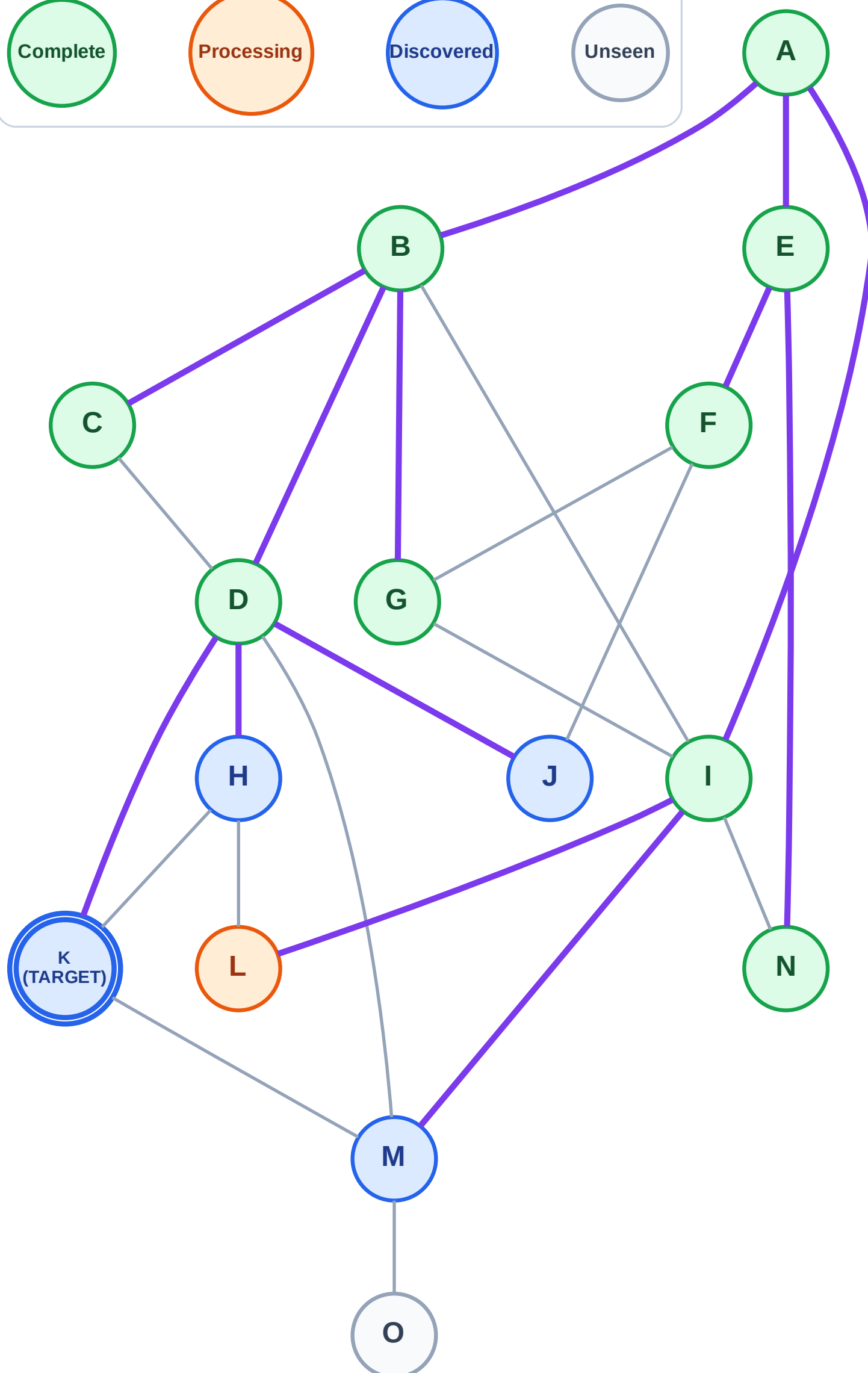
Discovered



Unseen

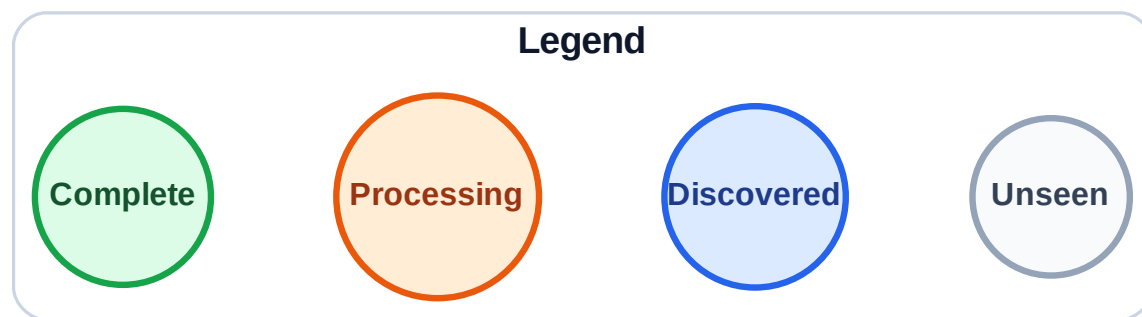
Queue: M → H → J → K

Visited: A, B, C, D, E, F, G, I, N



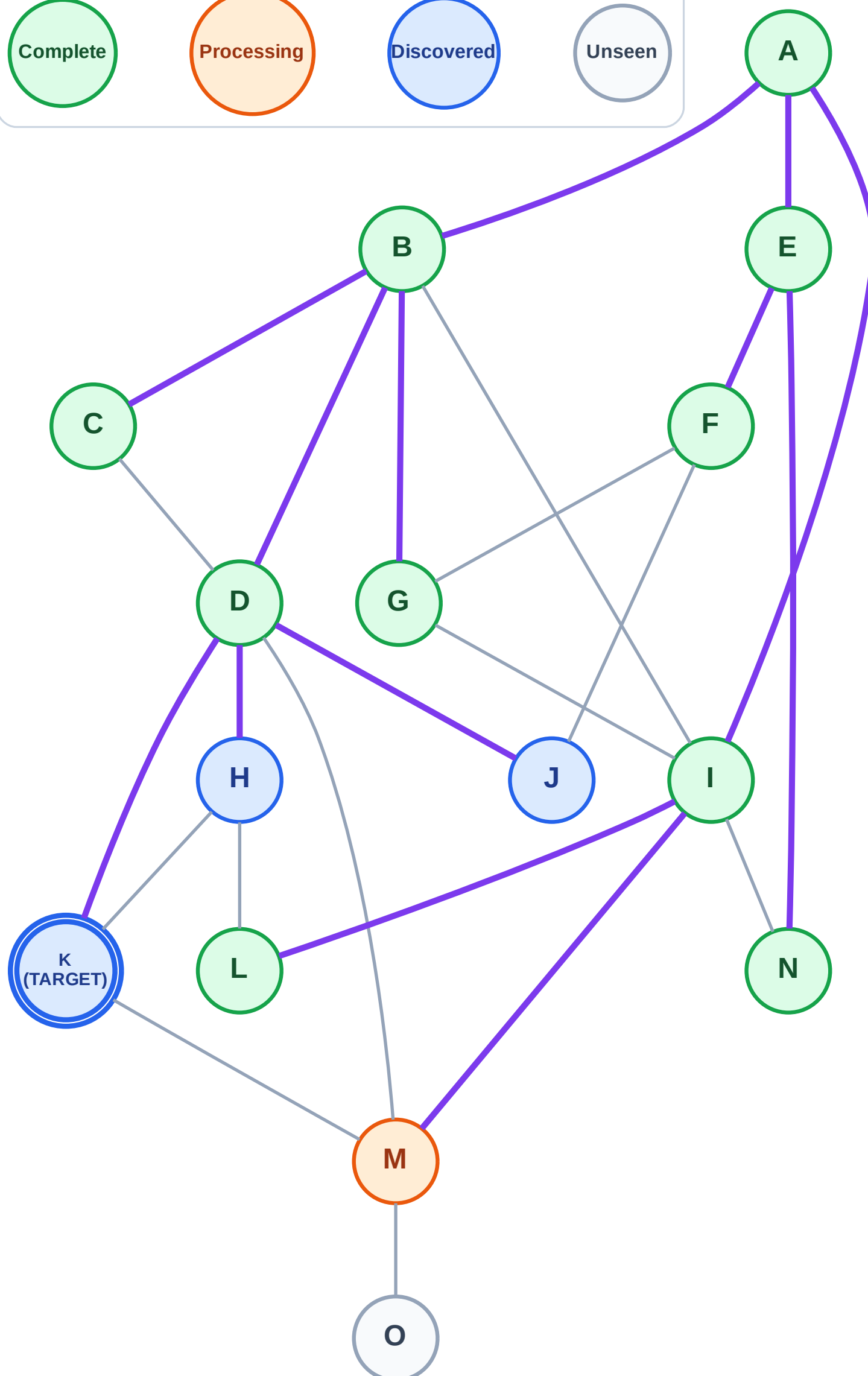
BFS Traversal

Step 22: Process node M



Queue: H → J → K

Visited: A, B, C, D, E, F, G, I, L, N



BFS Traversal

Step 23: Discover O from M

Legend

Complete

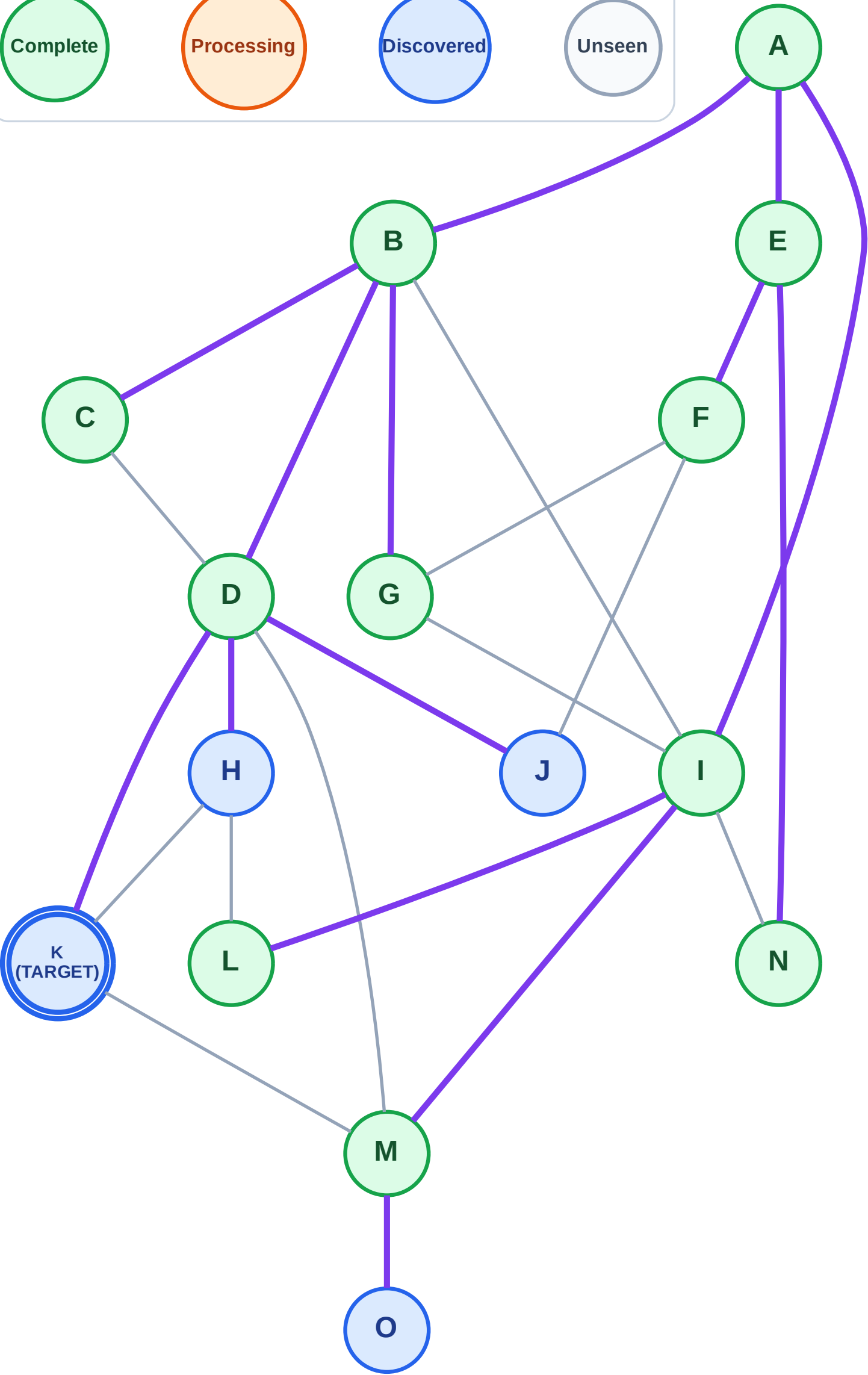
Processing

Discovered

Unseen

Queue: H → J → K → O

Visited: A, B, C, D, E, F, G, I, L, M, N



BFS Traversal

Step 24: Process node H

Legend

Complete

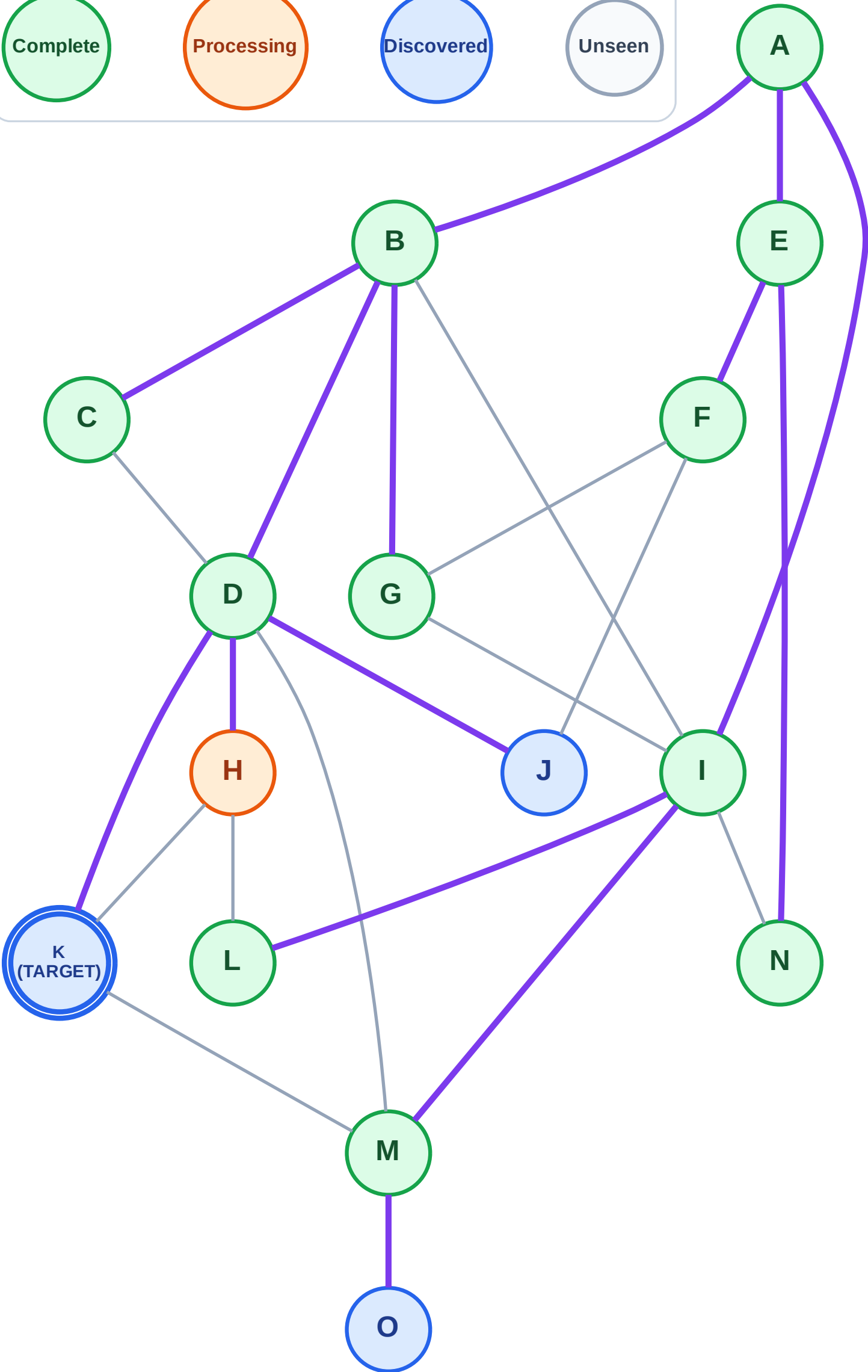
Processing

Discovered

Unseen

Queue: J → K → O

Visited: A, B, C, D, E, F, G, I, L, M, N



BFS Traversal

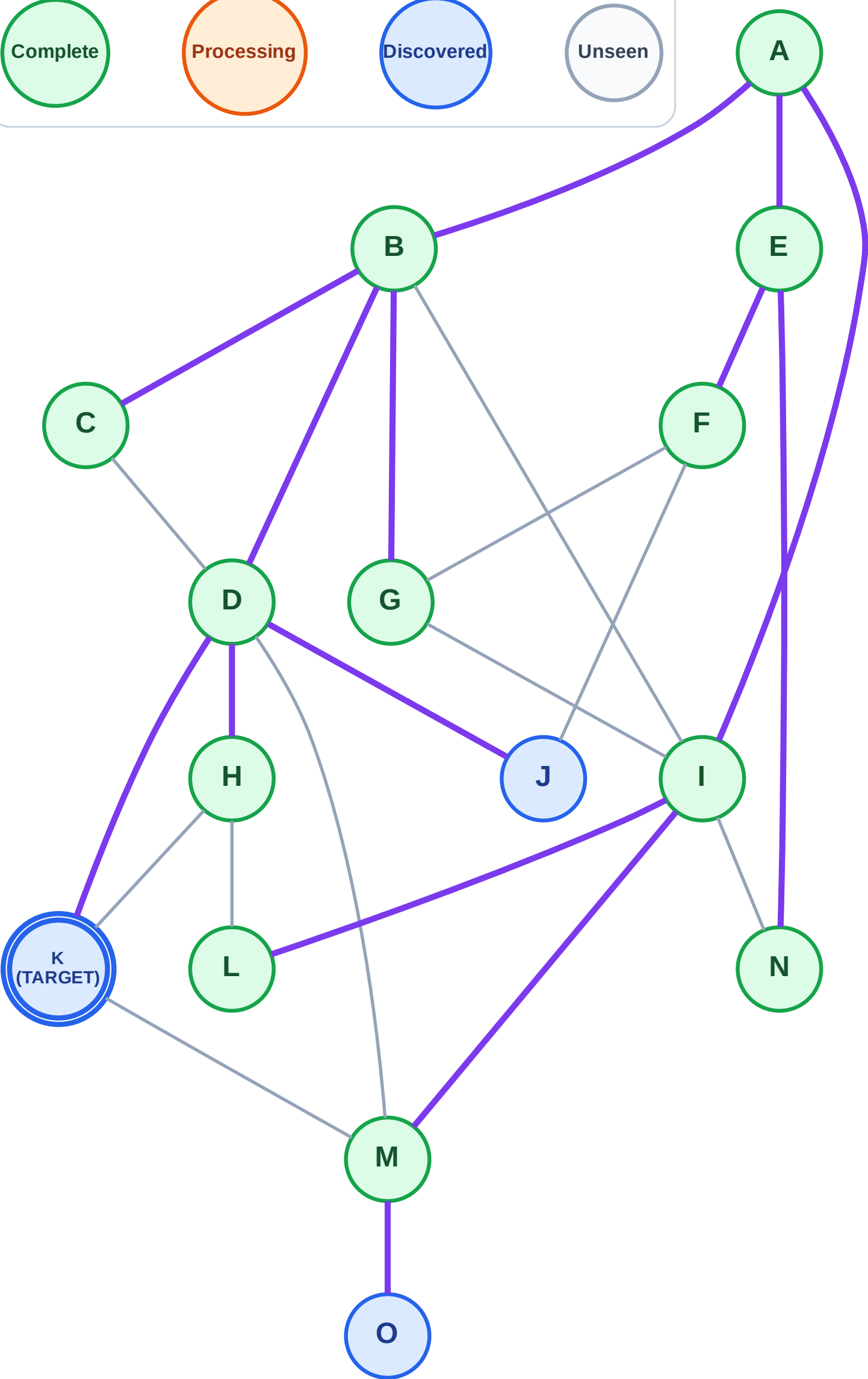
Step 25: No new neighbors from H

Legend



Queue: J → K → O

Visited: A, B, C, D, E, F, G, H, I, L, M, N



Step 26: Process node J



BFS Traversal

Step 27: No new neighbors from J

Legend



Complete



Processing



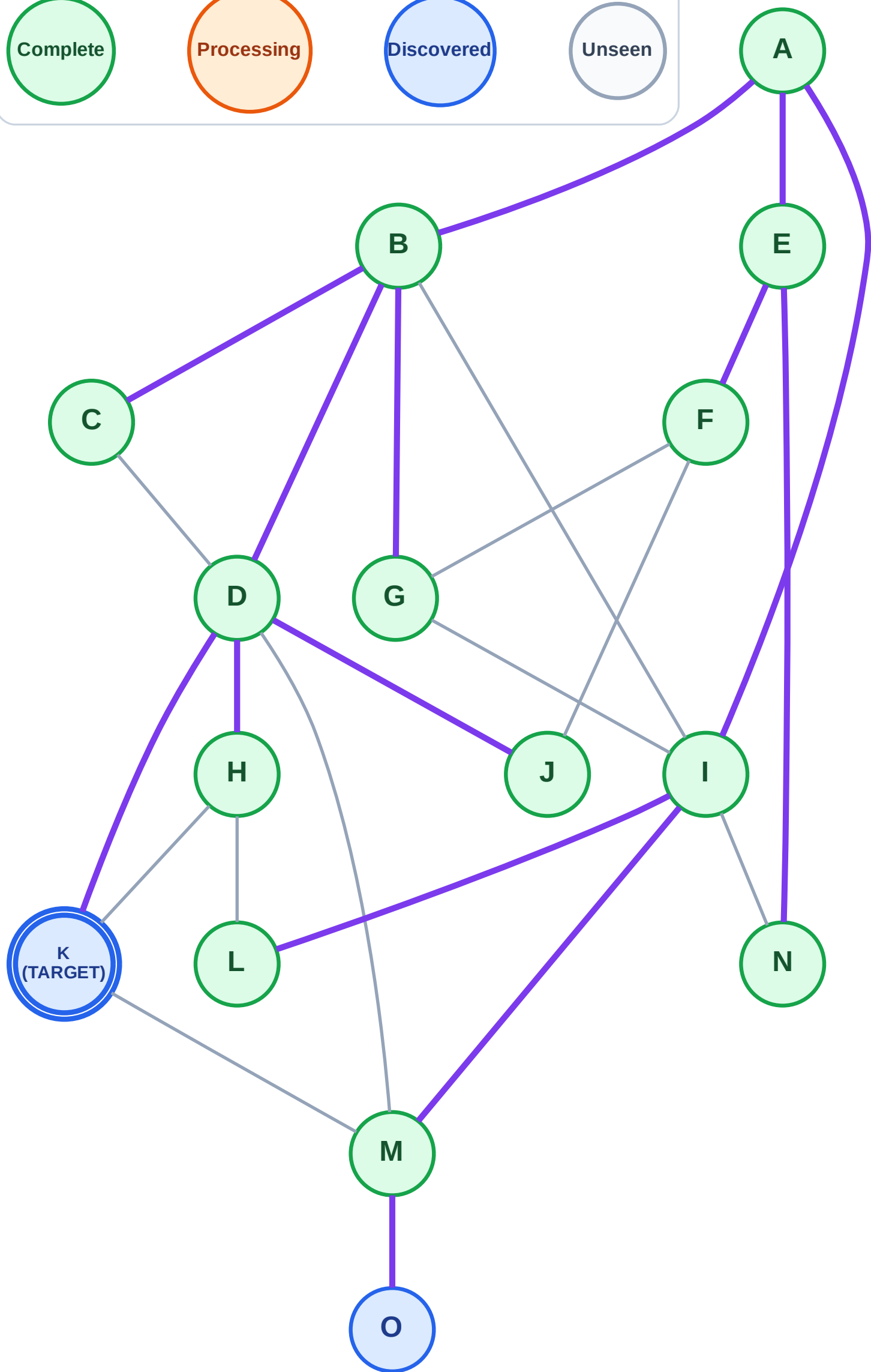
Discovered



Unseen

Queue: K → O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 28: Process node K

Legend

Complete

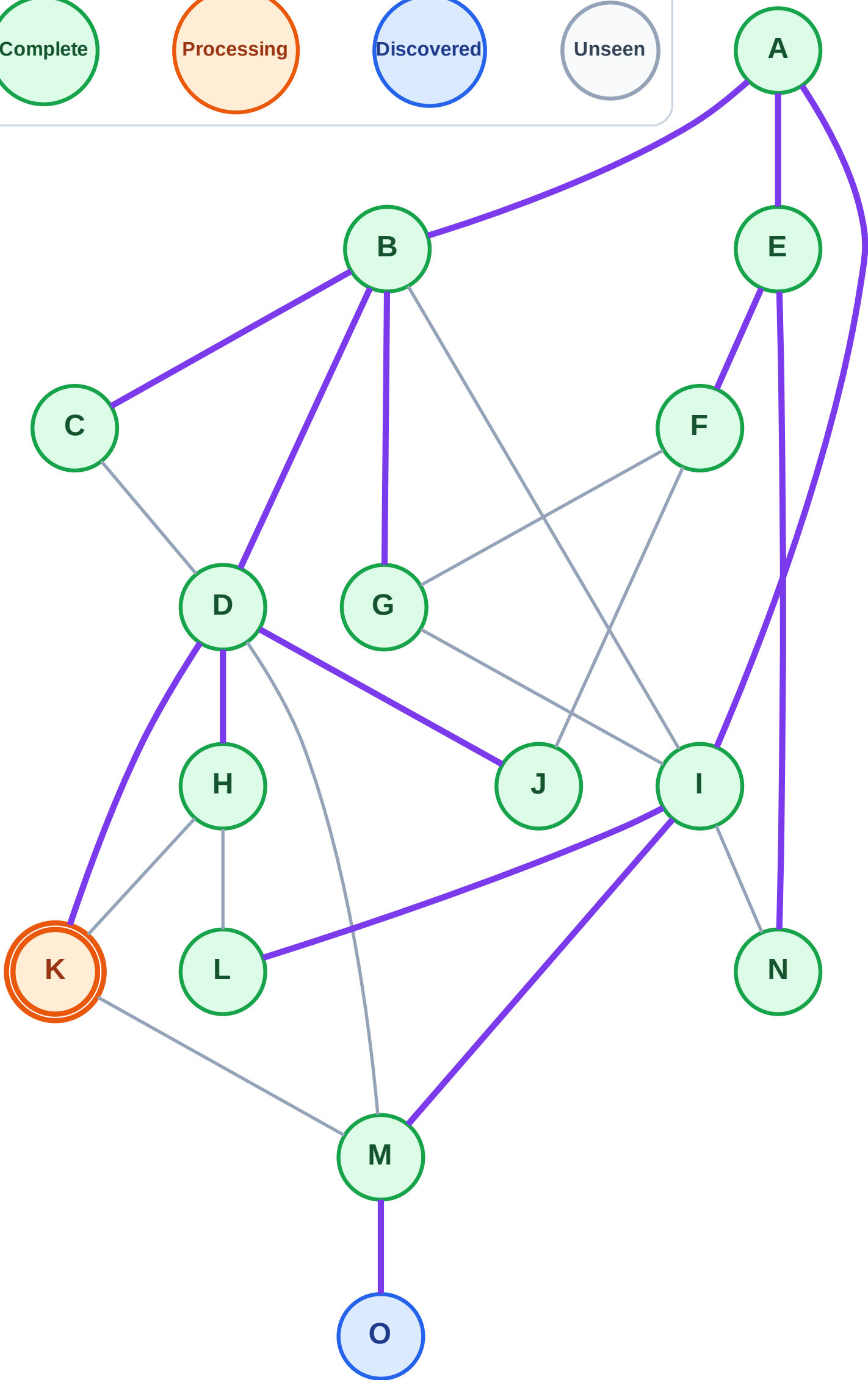
Processing

Discovered

Unseen

Queue: O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend

Complete

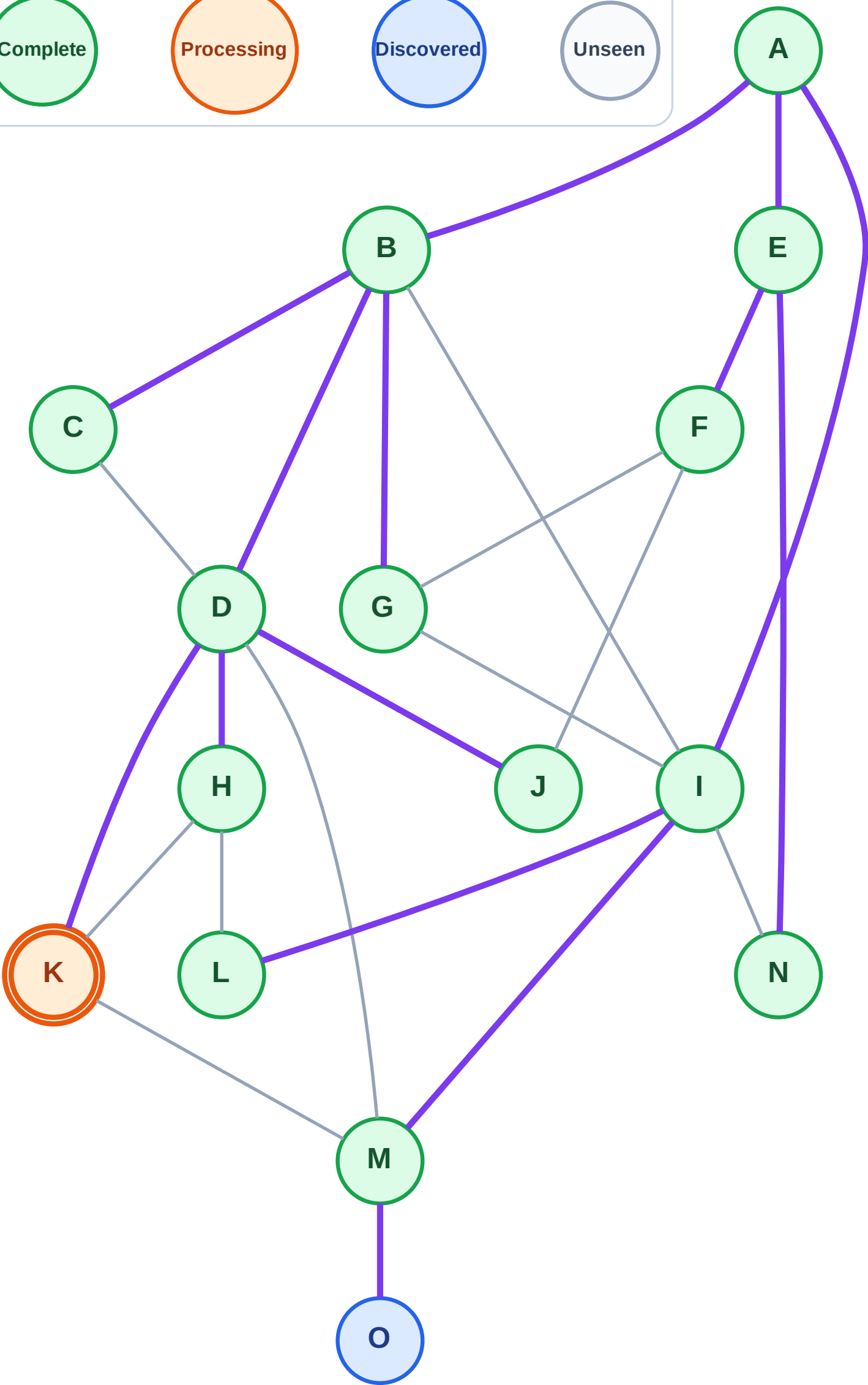
Processing

Discovered

Unseen

Queue: O

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

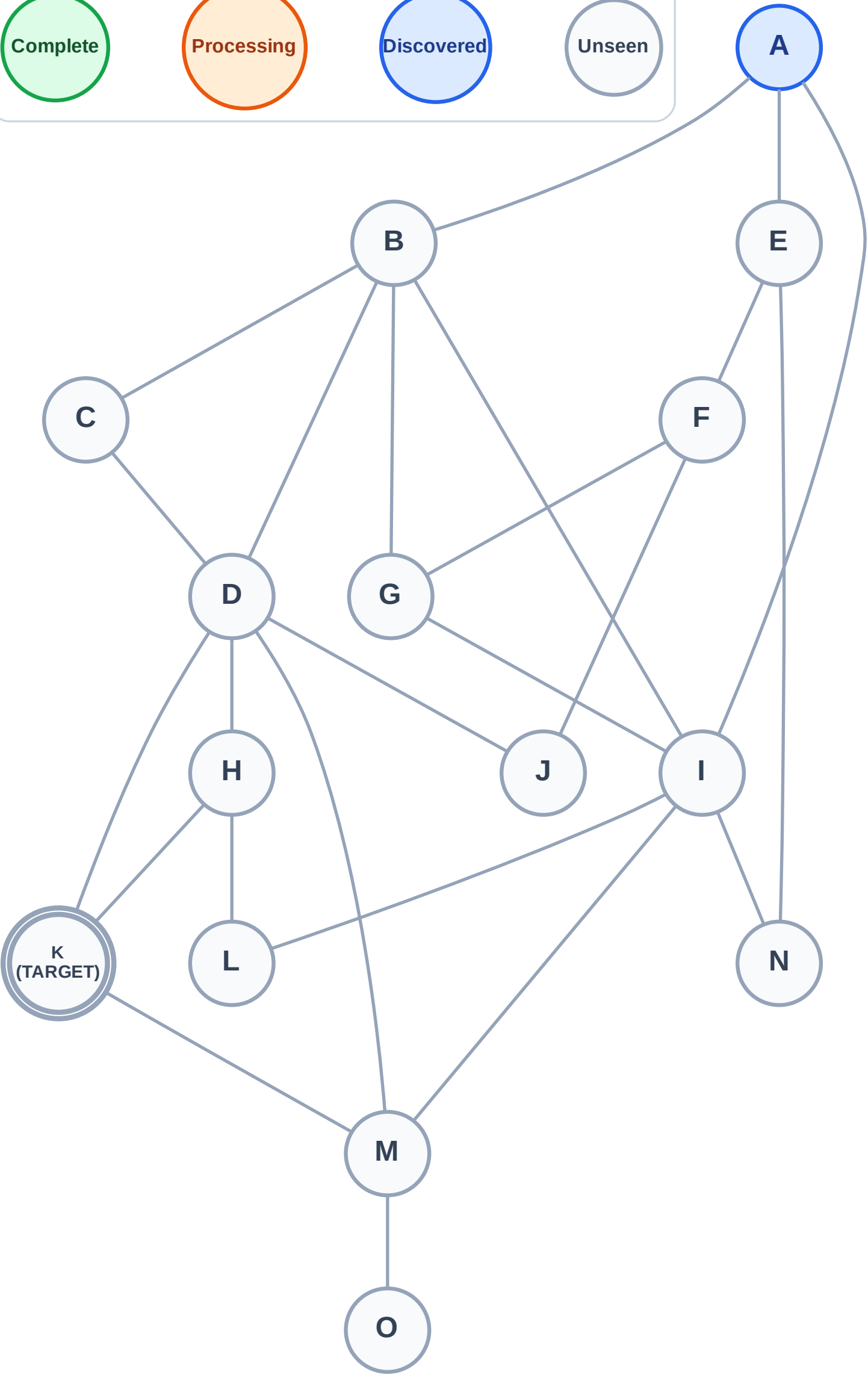
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

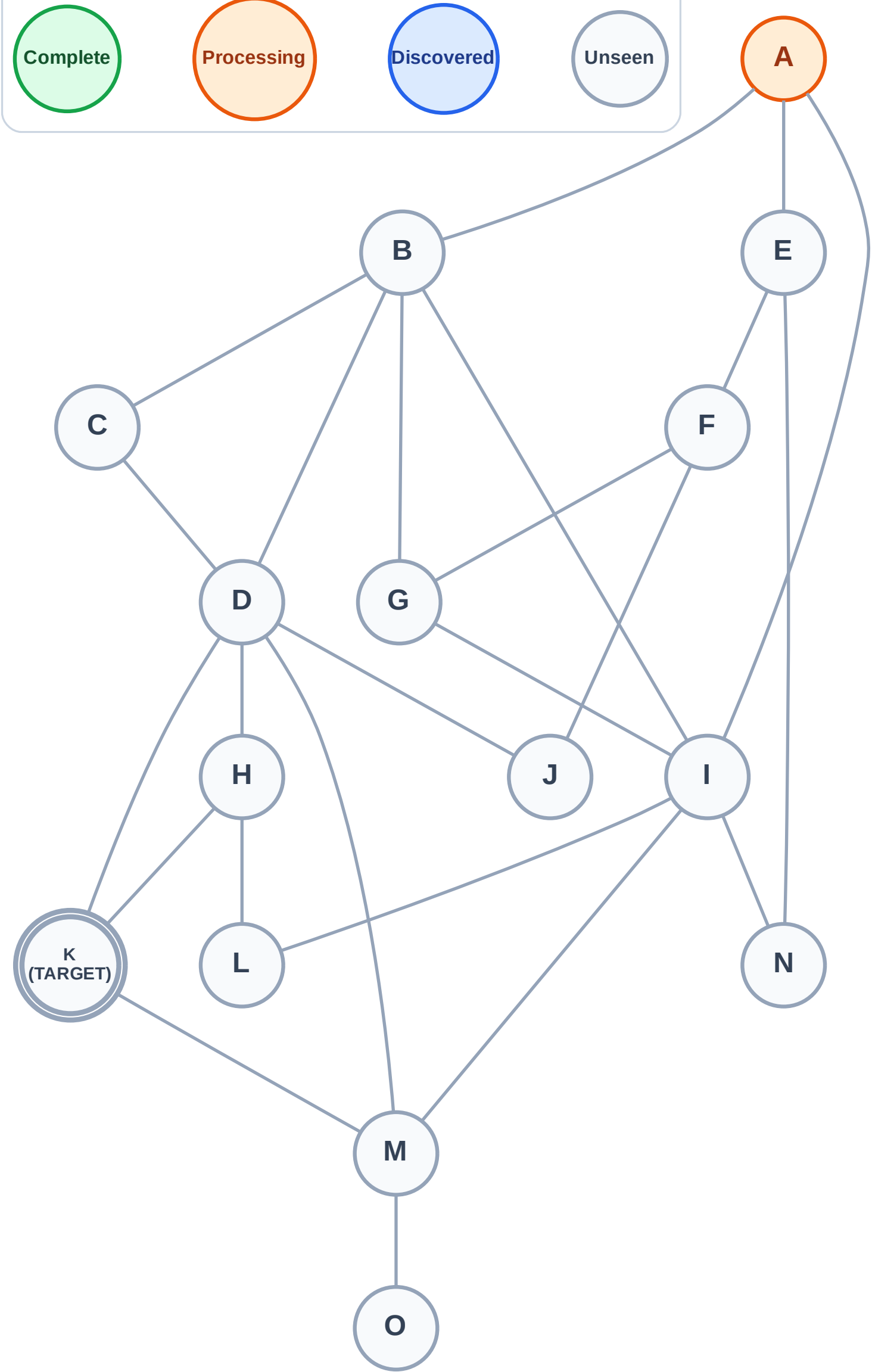
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

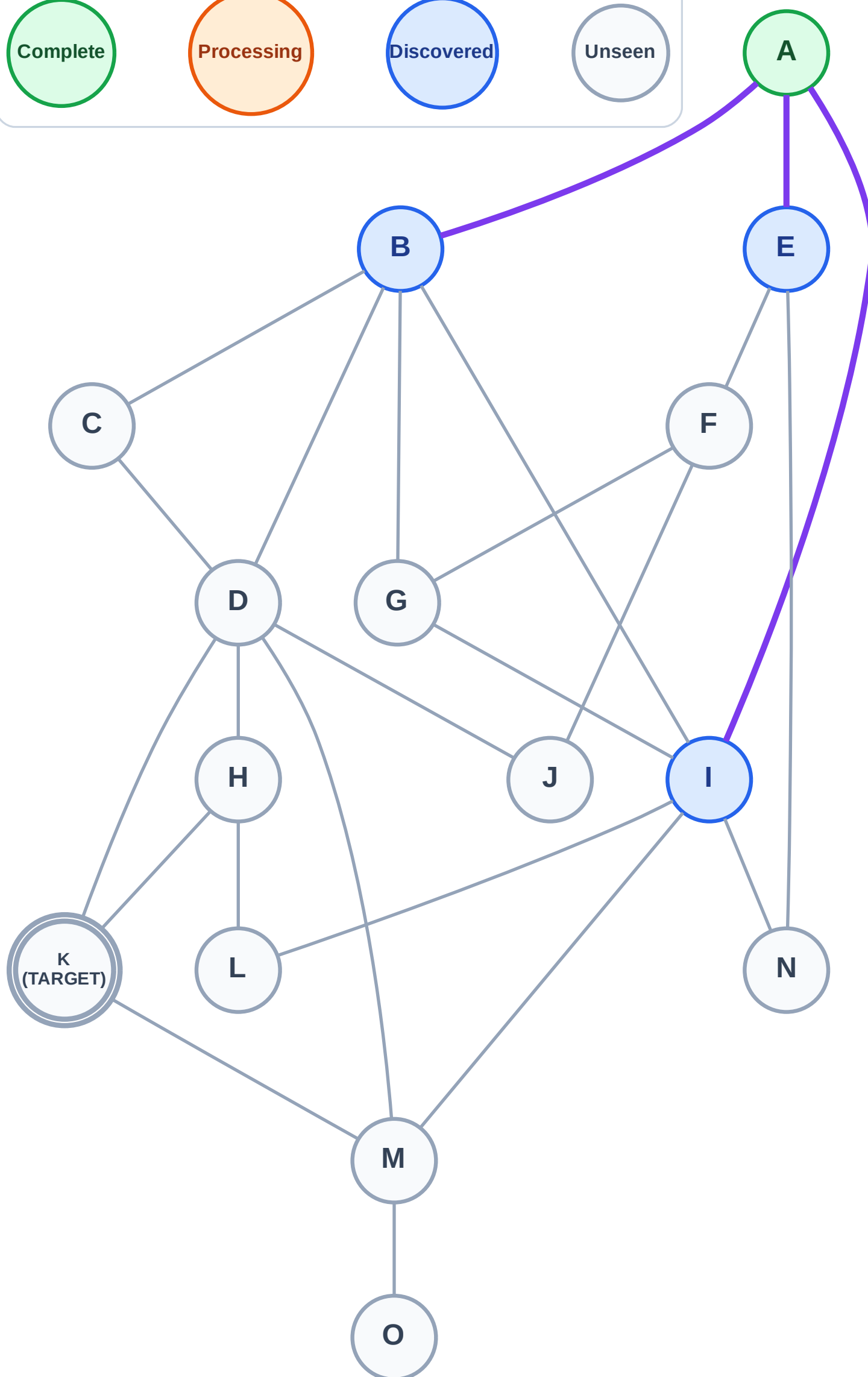
Step 3: Discover I, E, B from A

Legend



Stack (top at right): I | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

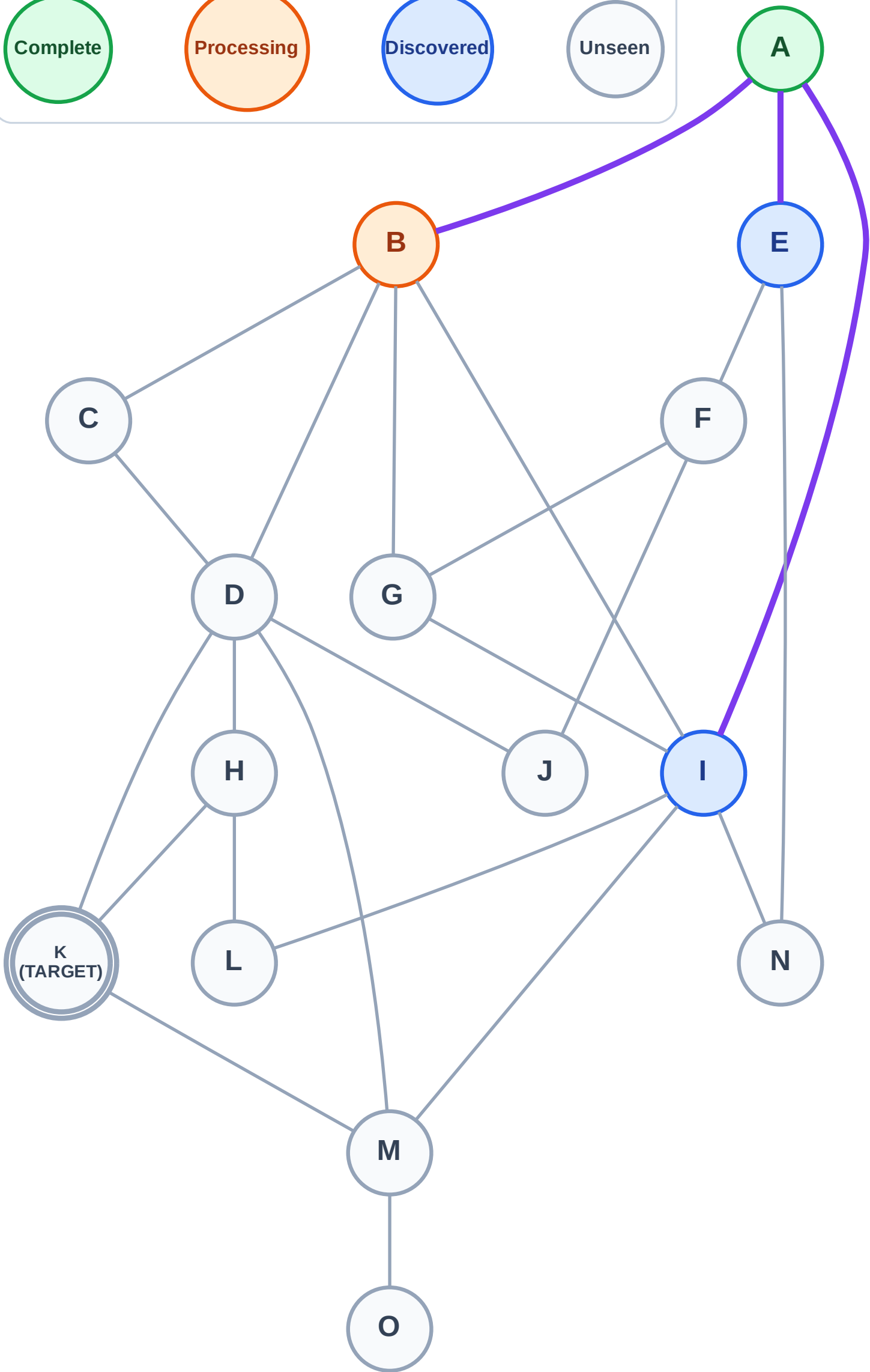
Processing

Discovered

Unseen

Stack (top at right): I | E

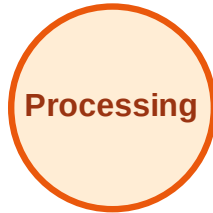
Visited: A



DFS Traversal

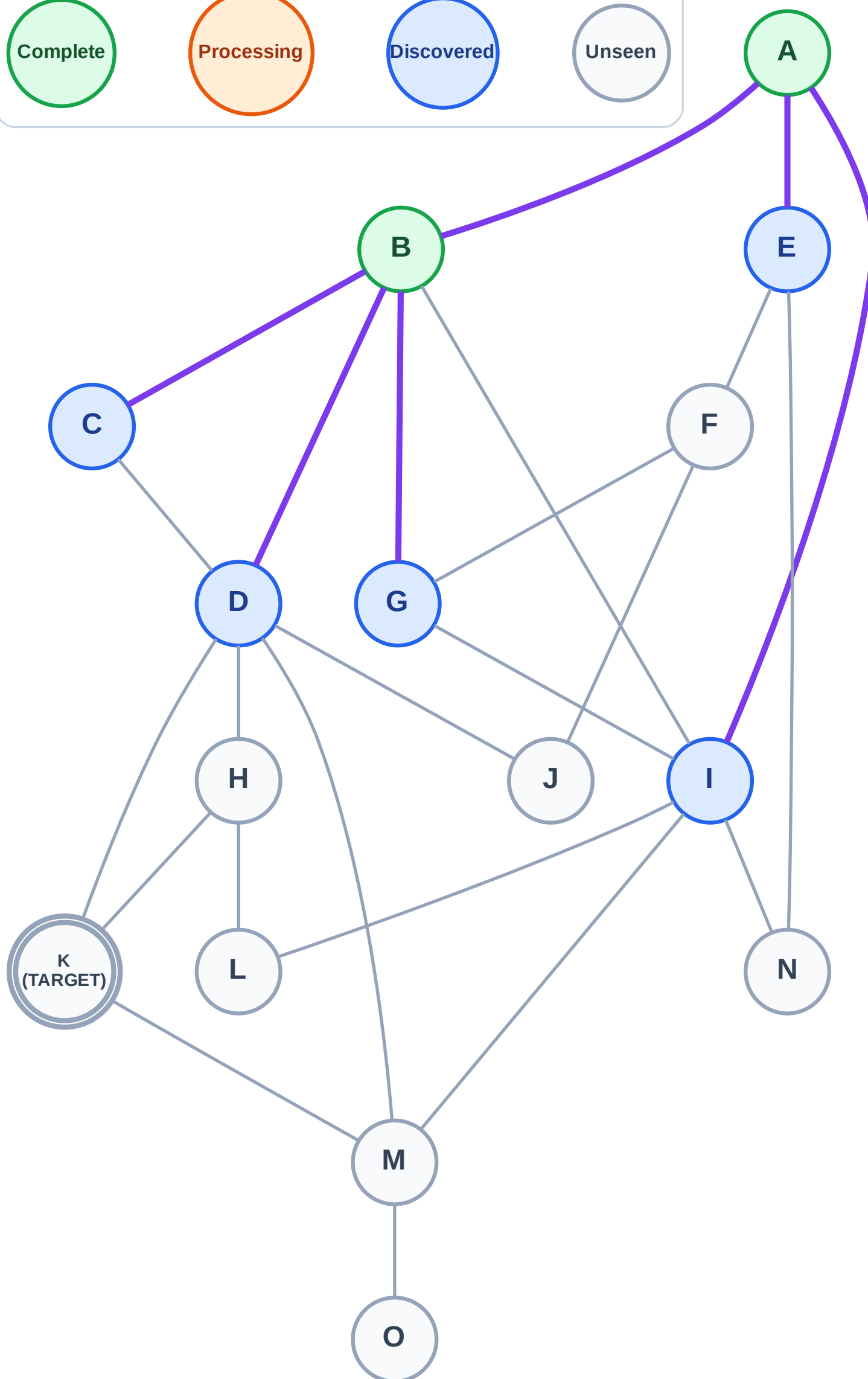
Step 5: Discover G, D, C from B

Legend



Stack (top at right): I | E | G | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



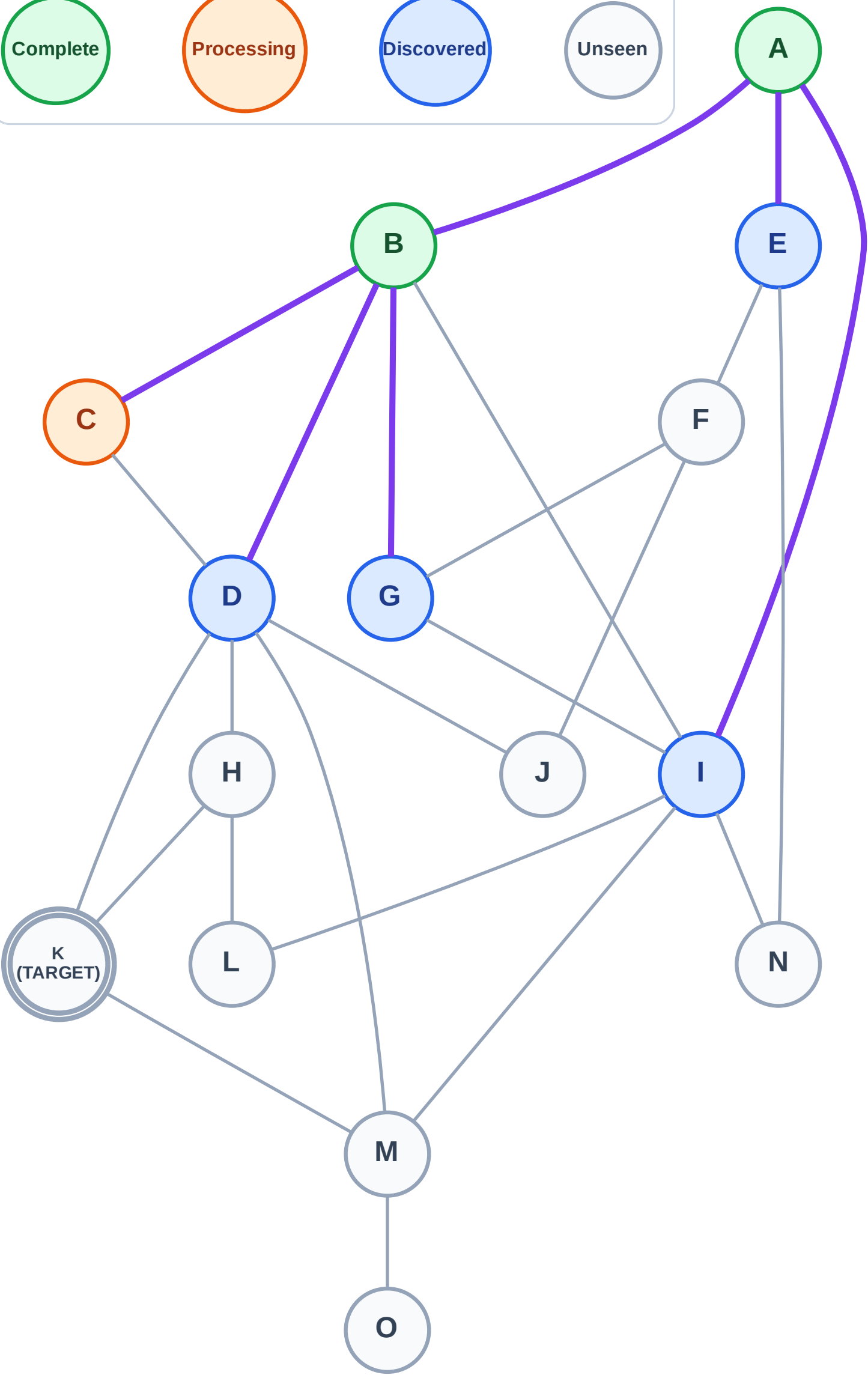
Discovered



Unseen

Stack (top at right): I | E | G | D

Visited: A, B



DFS Traversal

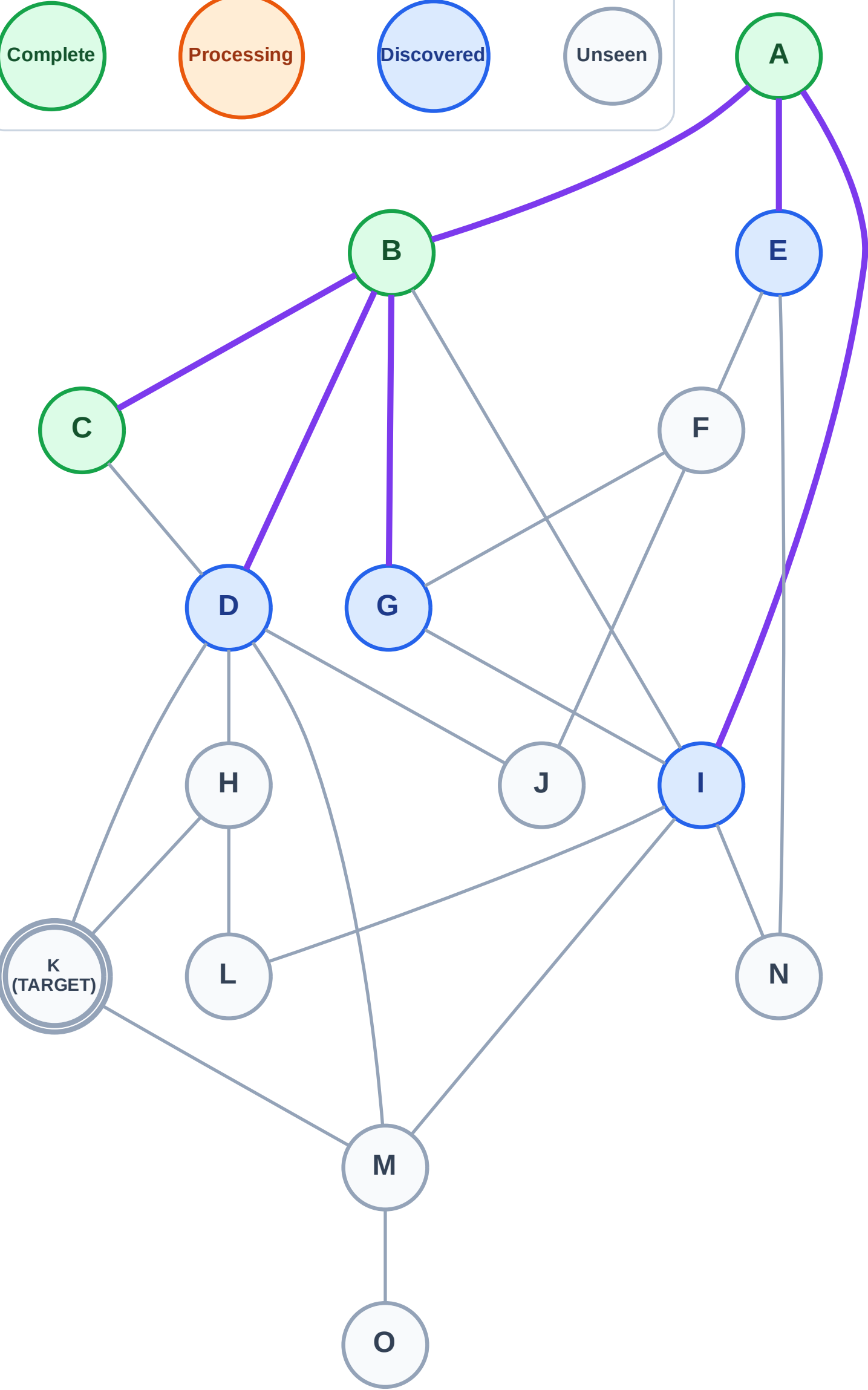
Step 7: No new neighbors from C

Legend



Stack (top at right): I | E | G | D

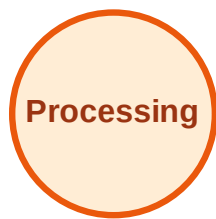
Visited: A, B, C



DFS Traversal

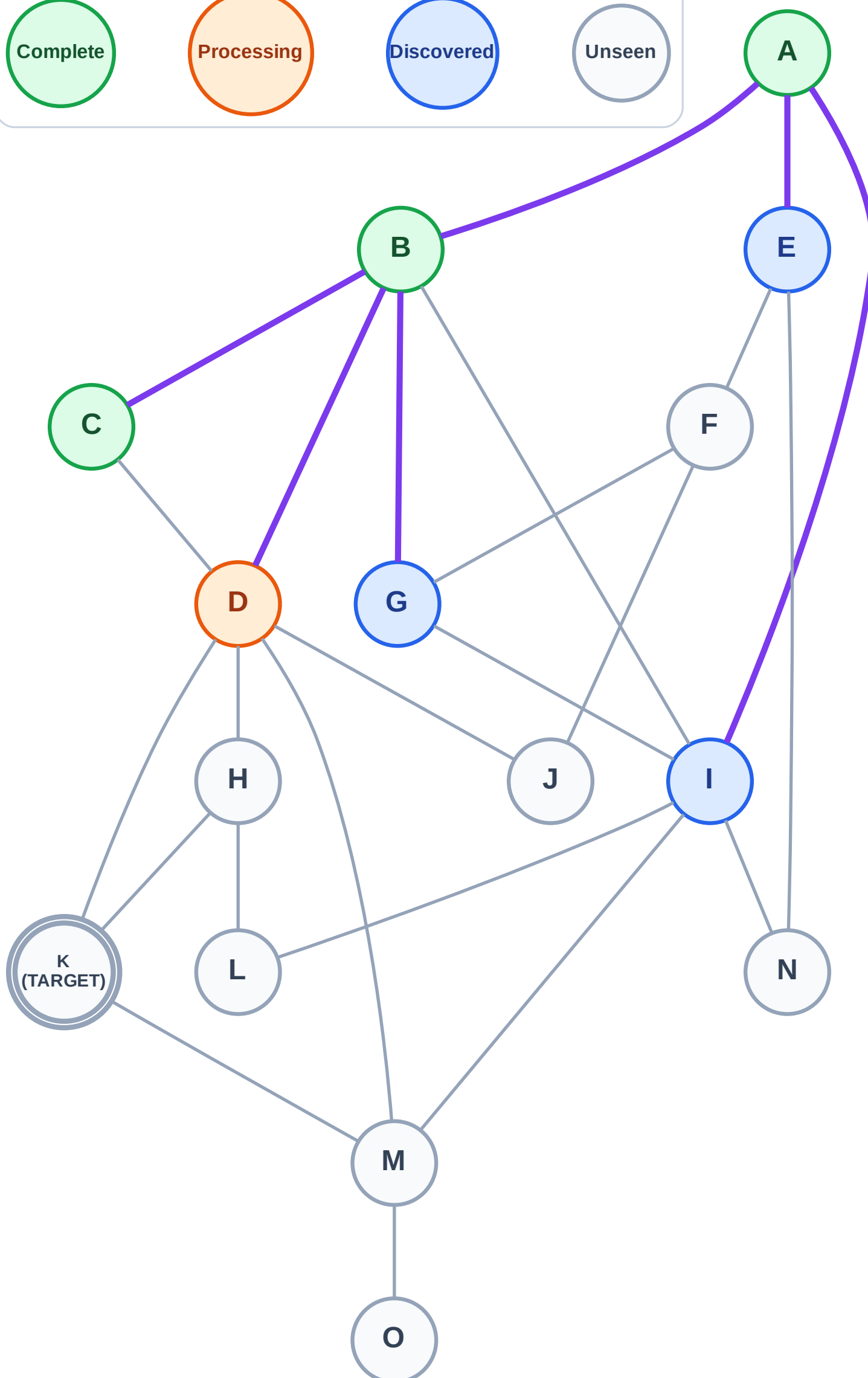
Step 8: Process node D

Legend



Stack (top at right): I | E | G

Visited: A, B, C



DFS Traversal

Step 9: Discover M, K, J, H from D

Legend

Complete

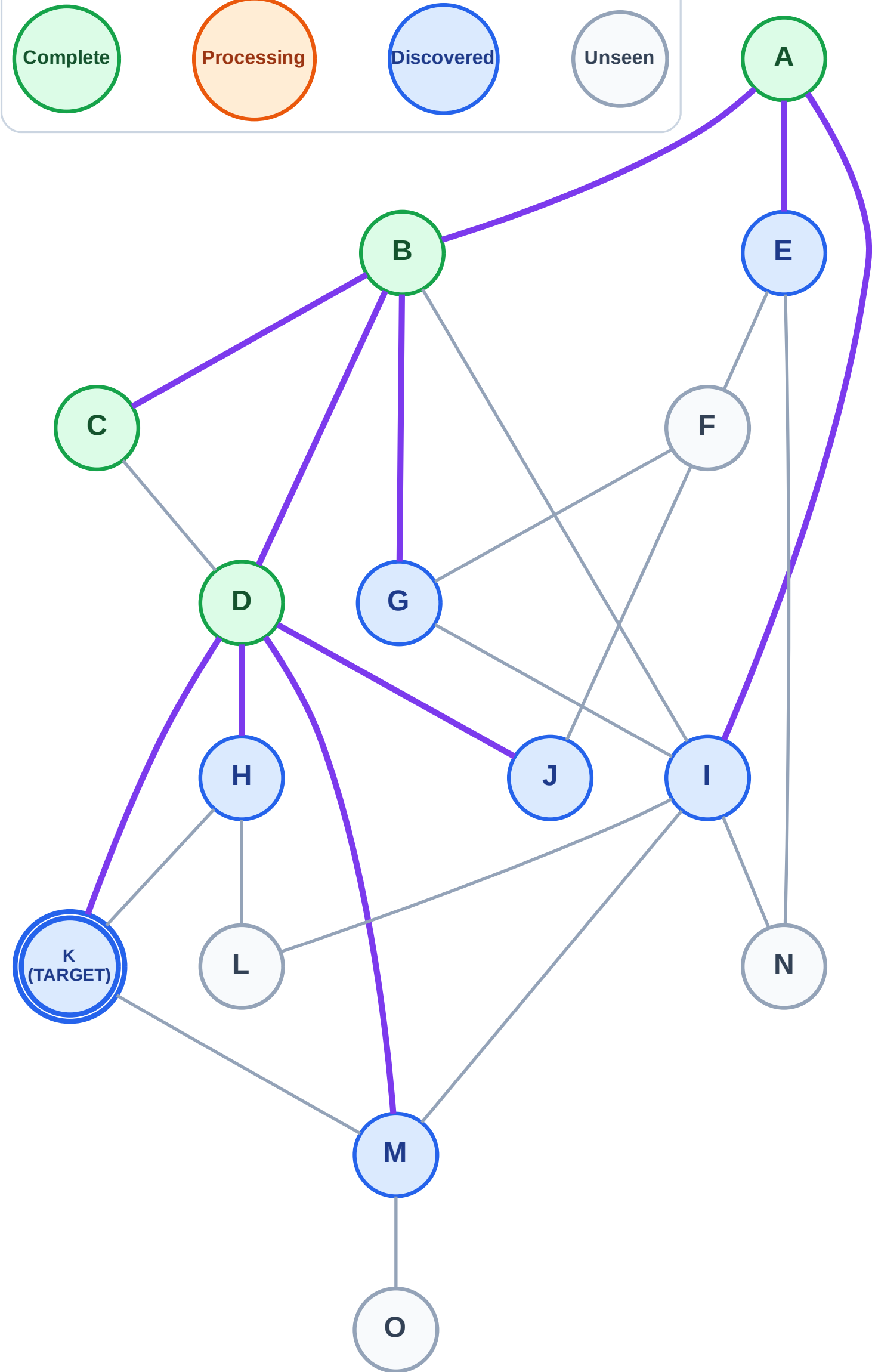
Processing

Discovered

Unseen

Stack (top at right): I | E | G | M | K | J | H

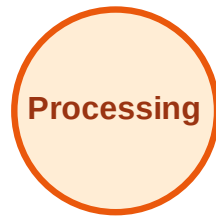
Visited: A, B, C, D



DFS Traversal

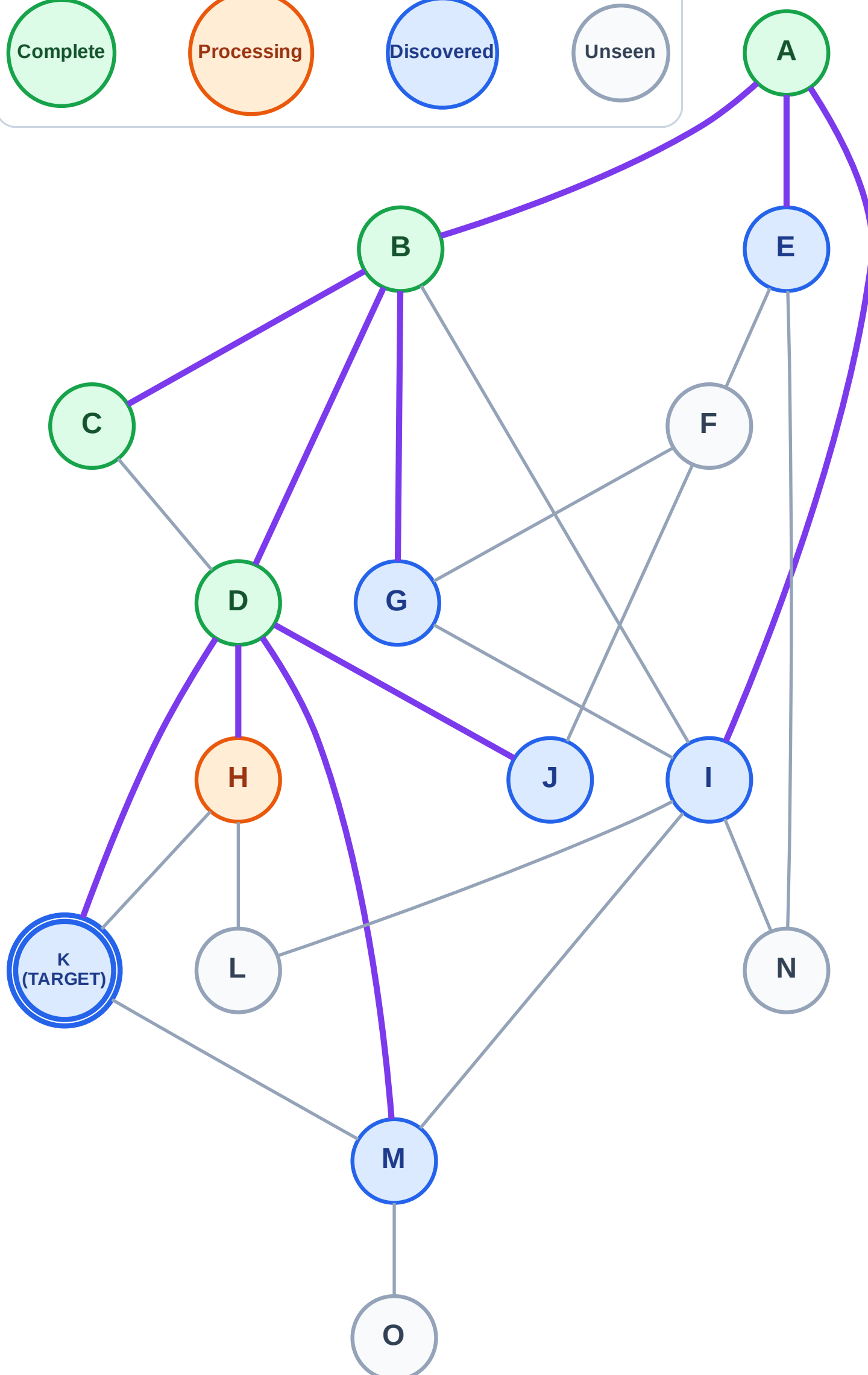
Step 10: Process node H

Legend



Stack (top at right): I | E | G | M | K | J

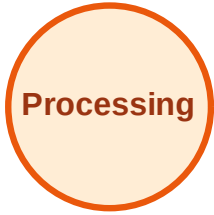
Visited: A, B, C, D



DFS Traversal

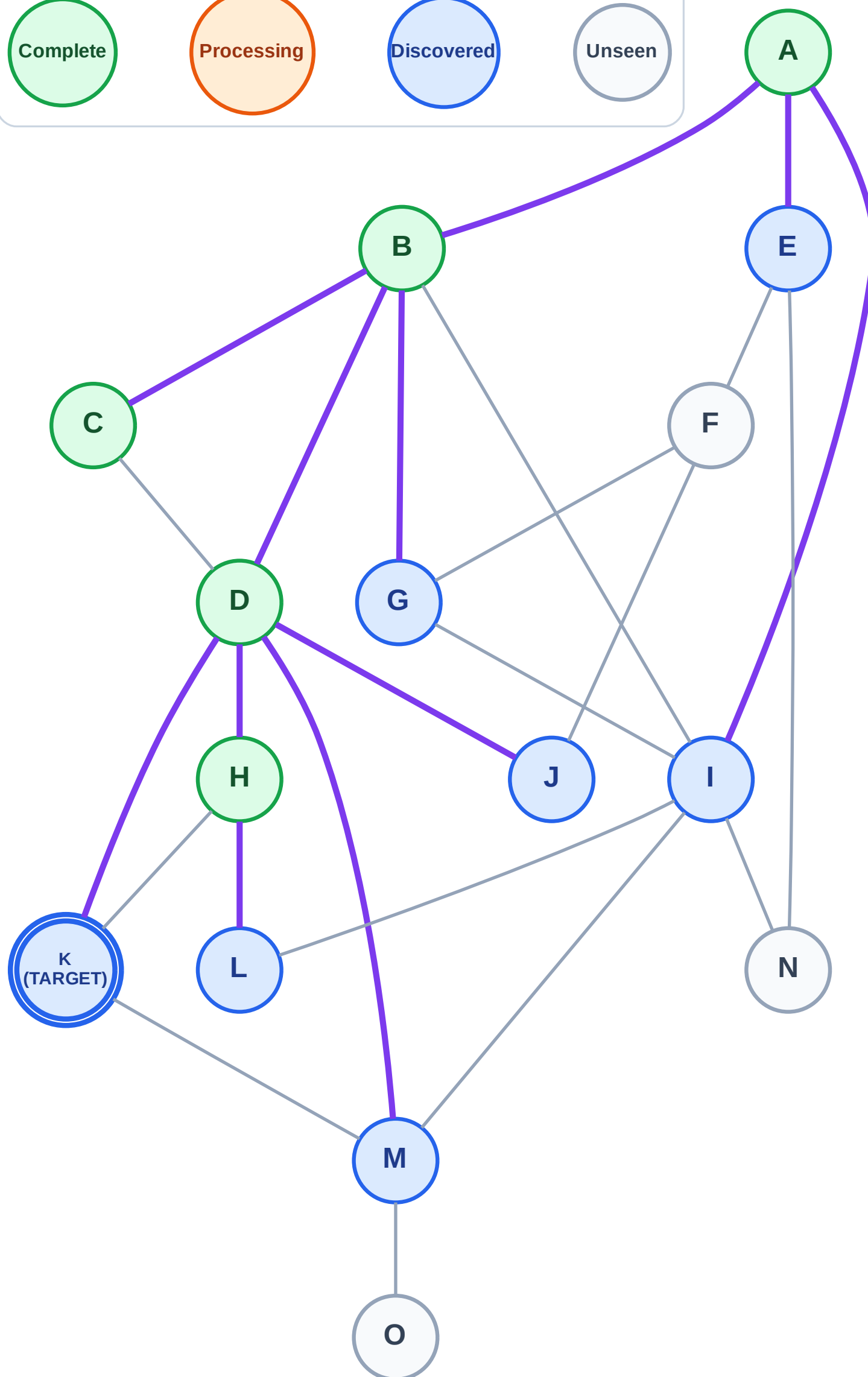
Step 11: Discover L from H

Legend



Stack (top at right): I | E | G | M | K | J | L

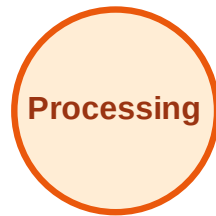
Visited: A, B, C, D, H



DFS Traversal

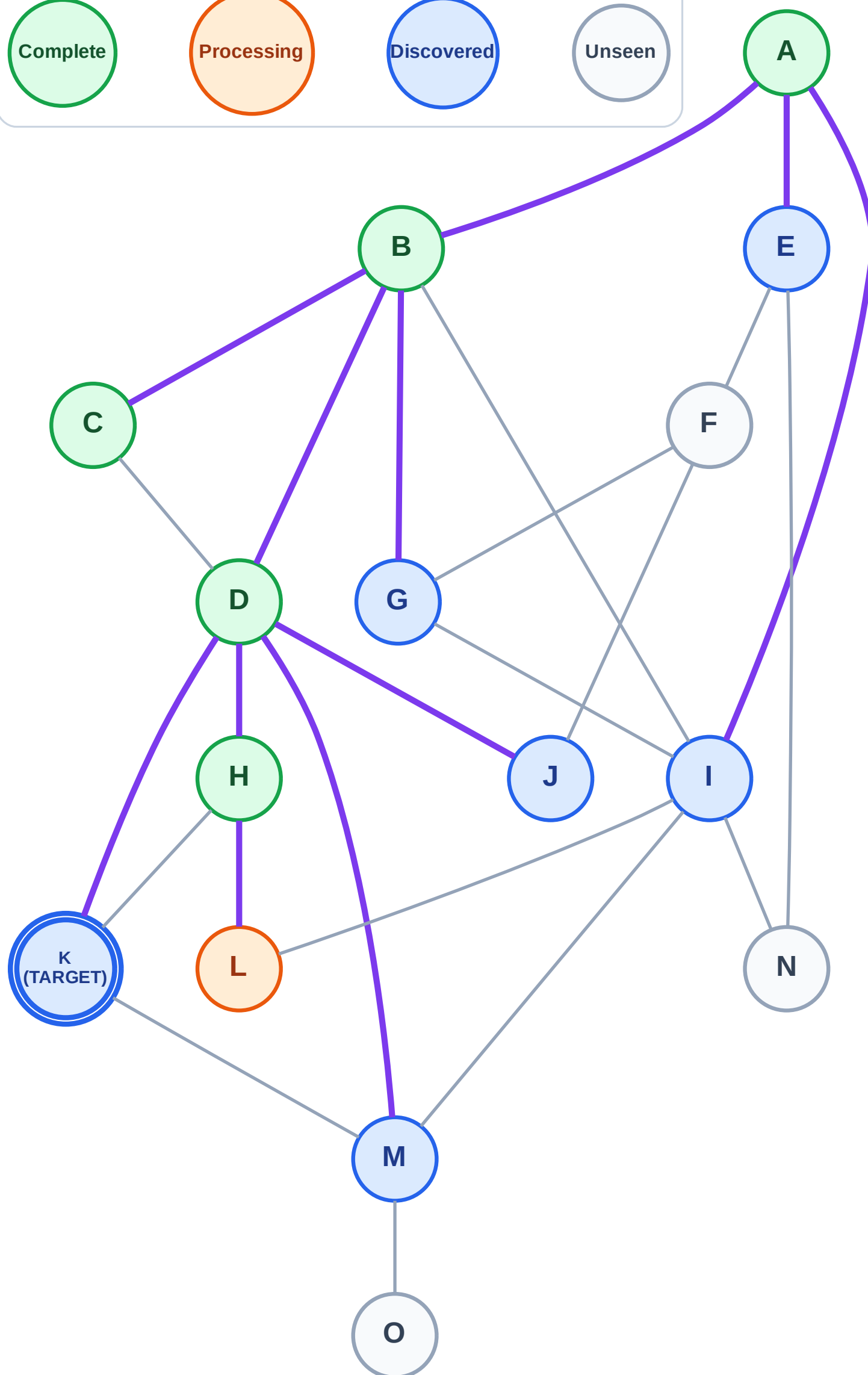
Step 12: Process node L

Legend



Stack (top at right): I | E | G | M | K | J

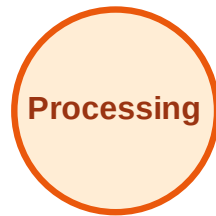
Visited: A, B, C, D, H



DFS Traversal

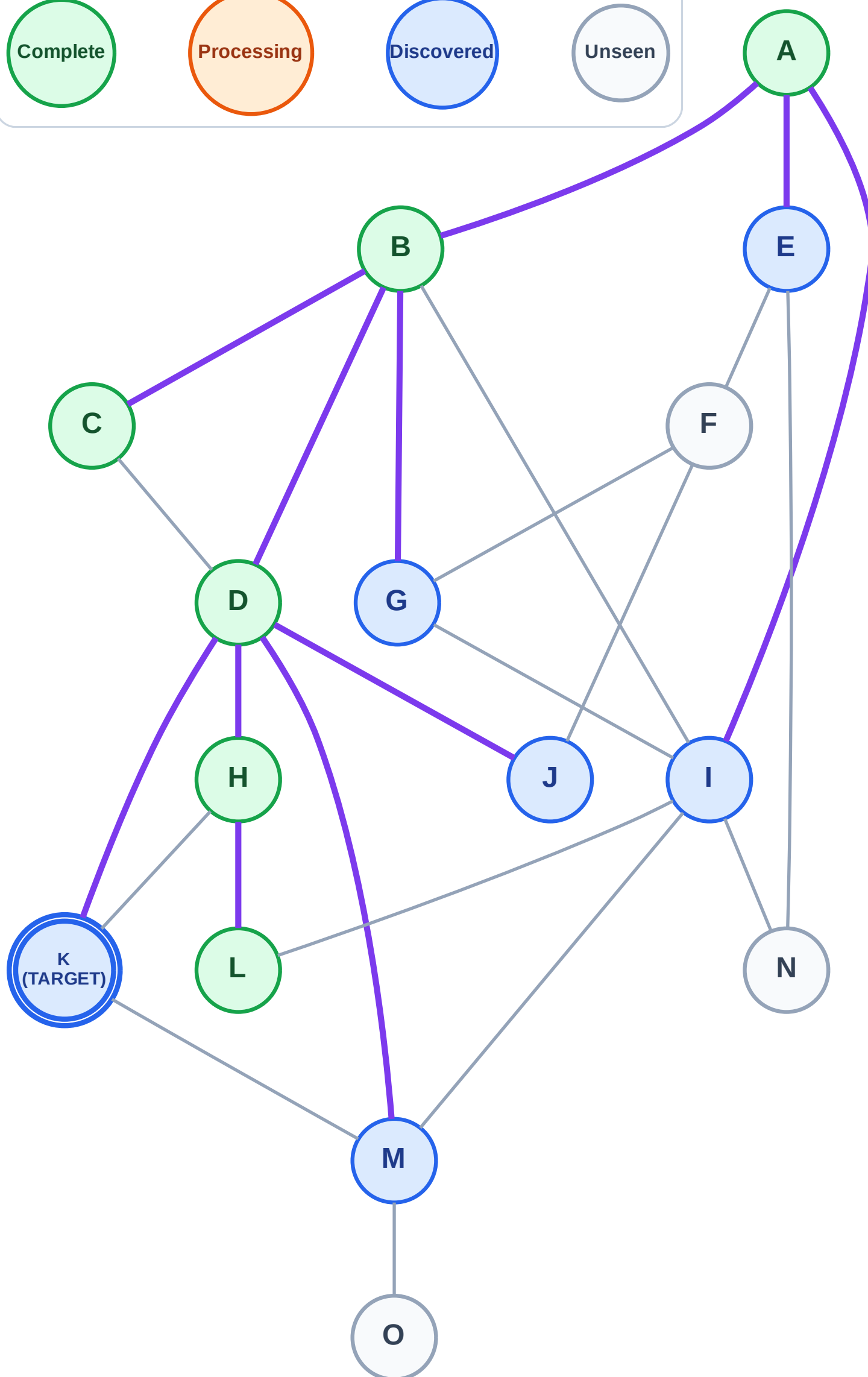
Step 13: No new neighbors from L

Legend



Stack (top at right): I | E | G | M | K | J

Visited: A, B, C, D, H, L



DFS Traversal

Step 14: Process node J

Legend

Complete

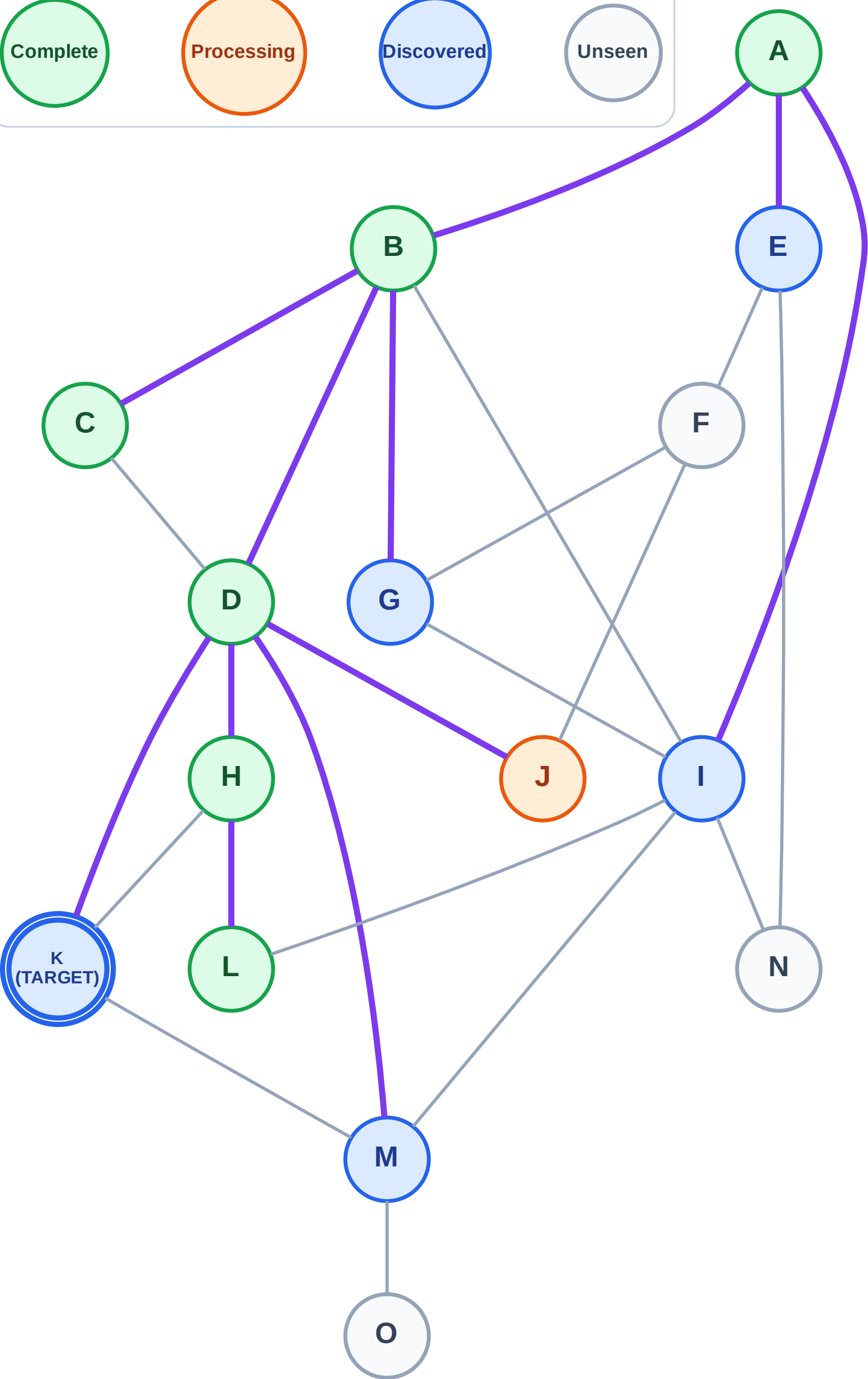
Processing

Discovered

Unseen

Stack (top at right): I | E | G | M | K

Visited: A, B, C, D, H, L

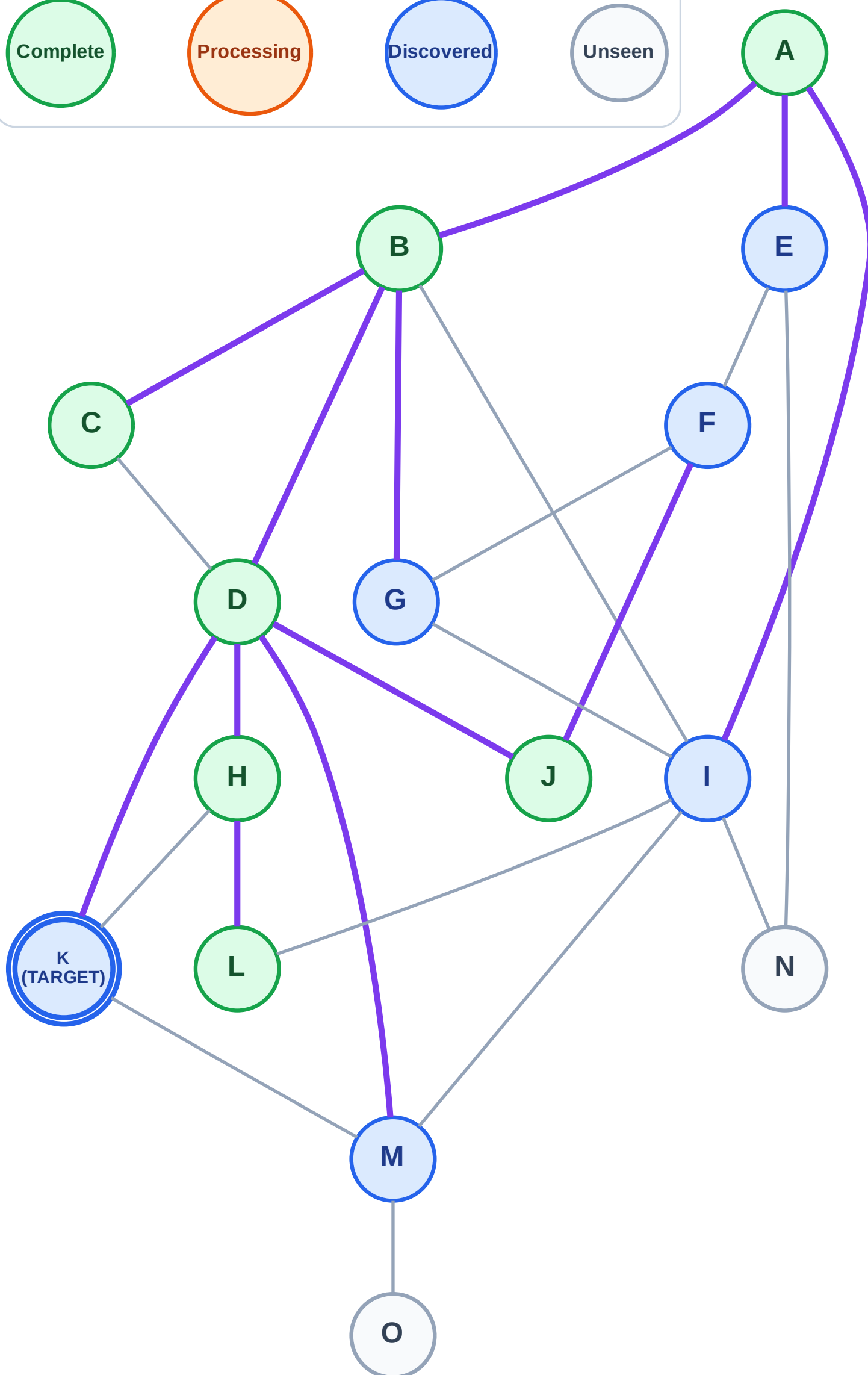


Step 15: Discover F from J

Legend



Visited: A, B, C, D, H, J, L



DFS Traversal

Step 16: Process node F

Legend

Complete

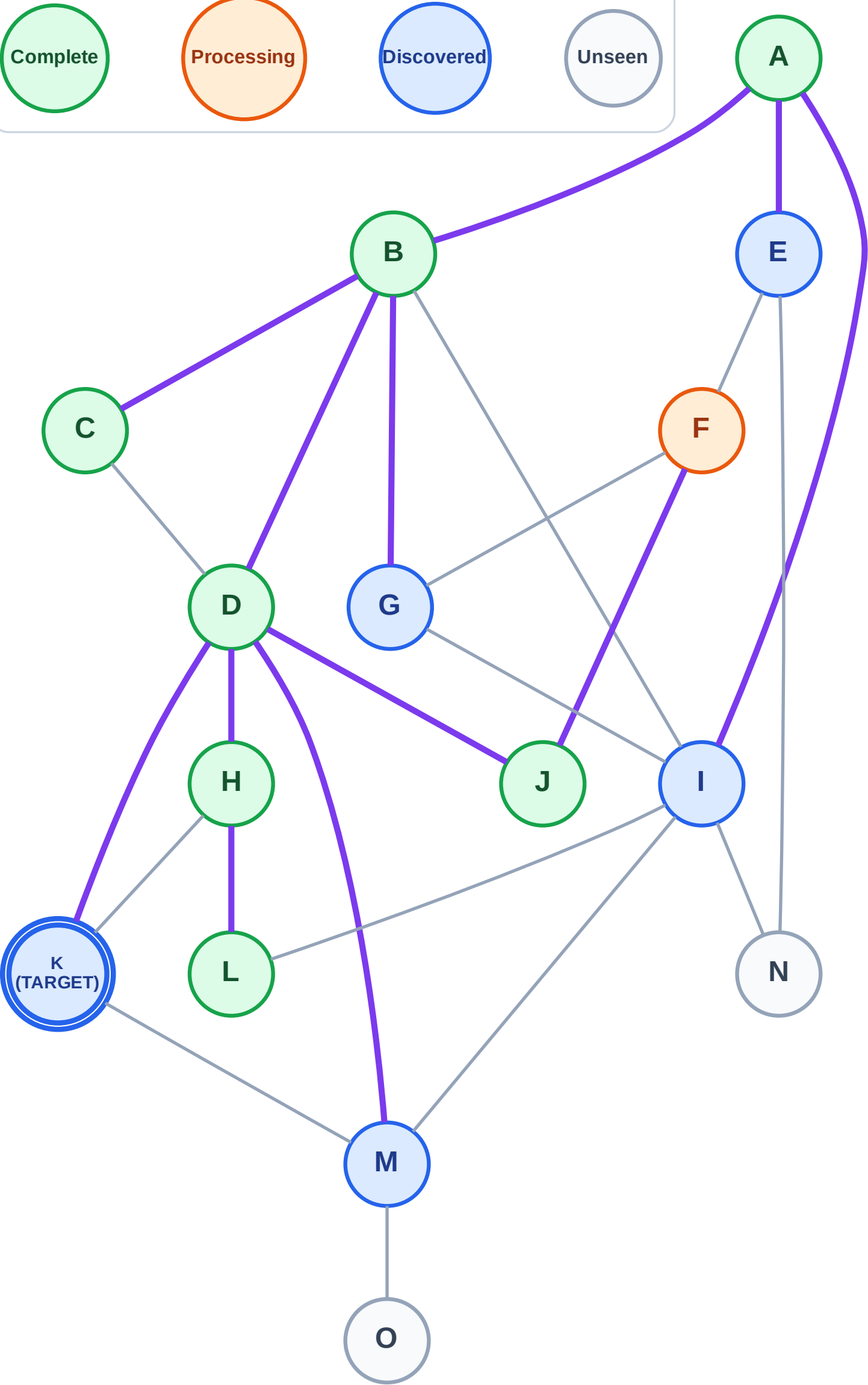
Processing

Discovered

Unseen

Stack (top at right): I | E | G | M | K

Visited: A, B, C, D, H, J, L



DFS Traversal

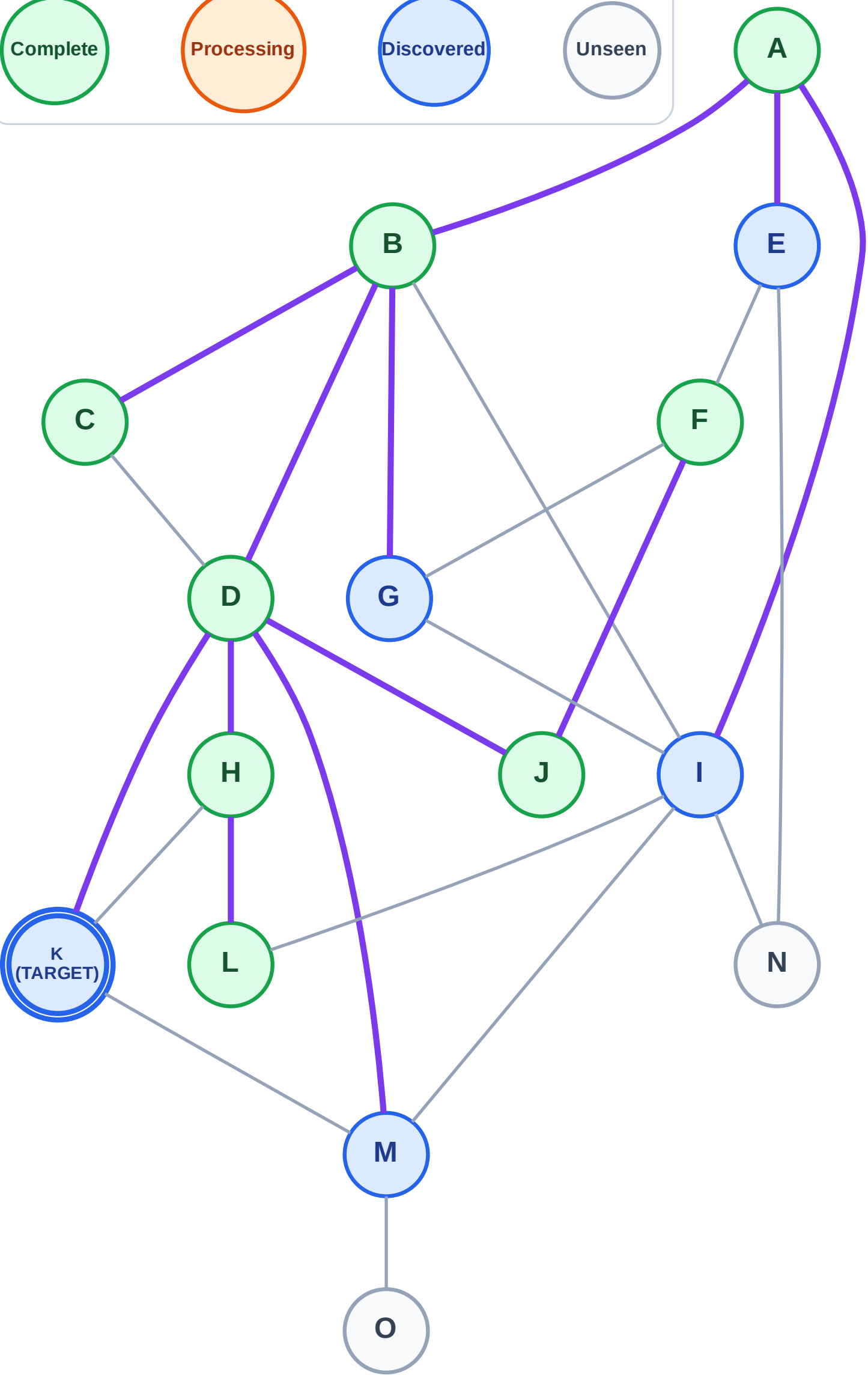
Step 17: No new neighbors from F

Legend



Stack (top at right): I | E | G | M | K

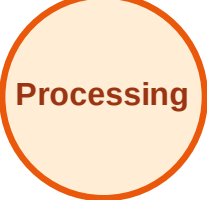
Visited: A, B, C, D, F, H, J, L



DFS Traversal

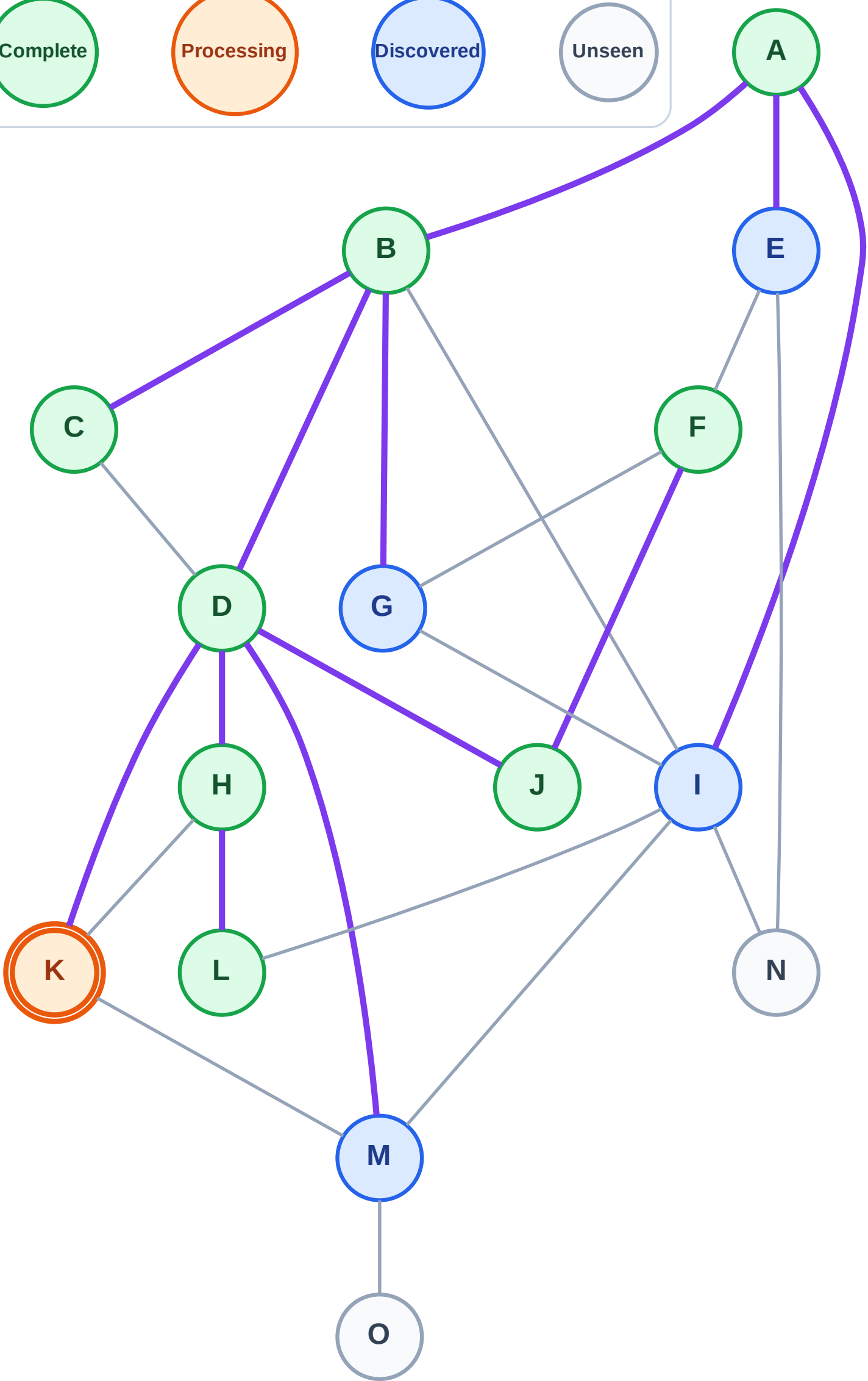
Step 18: Process node K

Legend



Stack (top at right): I | E | G | M

Visited: A, B, C, D, F, H, J, L



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend



Stack (top at right): I | E | G | M

Visited: A, B, C, D, F, H, J, K, L

